

ADVERSARIAL ATTACKS LEVERAGE INTERFERENCE BETWEEN FEATURES IN SUPERPOSITION

Anonymous authors
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ABSTRACT

Fundamental questions persist concerning when, and for what mechanistic reasons, adversarial examples (AExs) arise in neural networks (NNs), with prevailing accounts alternately construing them as artifacts of irregular decision landscapes or as consequences of sensitivity to non-robust input features. In this paper, we advance a complementary account: adversarial vulnerability can emerge from *efficient* information encoding in NNs. More precisely, we show that superposition – the representational regime in which networks encode more features than they possess dimensions – induces latent arrangements whose interference structure can be systematically exploited by adversaries. We demonstrate that adversarial perturbations leverage interference among superposed features, rendering attack patterns predictable from the geometry of feature arrangements. The resulting framework offers a mechanistic explanation for two well-documented phenomena: the transferability of adversarial attacks across models trained under similar regimes and the emergence of class-specific vulnerability patterns. In synthetic settings where superposition is precisely controlled, we establish that superposition *suffices* to generate adversarial vulnerability. We then show that the same qualitative findings persist in a vision transformer (ViT) trained on CIFAR-10. Taken together, these results indicate that adversarial vulnerability can be a byproduct of representational compression, rather than merely a defect of the learning procedure or a property of non-robust inputs. We release our implementation through a software repository.

1 INTRODUCTION

Despite extensive investigation of AExs (Szegedy et al., 2014; Goodfellow et al., 2014), their fundamental origin remains unsettled. In particular, we still lack a sufficiently mechanistic account of how ostensibly minimal perturbations can so dramatically alter model predictions, and this explanatory gap correspondingly limits our ability to anticipate which perturbations will succeed, to explain why attacks transfer across models, or to derive principled defences rather than empirically tuned countermeasures.

Existing explanations can be organised, at a coarse level, into two broad families (Nakkiran, 2019). The *bug* perspective attributes vulnerability to factors such as deficiencies in learning (Schmidt et al., 2018; Nakkiran, 2019) or to geometric properties of high-dimensional decision boundaries (Fawzi et al., 2016); work in this tradition often supplies statistical bounds on quantities such as capacity (Bubeck & Sellke, 2021) or Lipschitz constraints (Hein & Andriushchenko, 2017), yet does not specify which concrete perturbations should succeed. By contrast, the *feature* perspective argues that AExs exploit predictive but non-robust statistical patterns in the data (Ilyas et al., 2019), thereby shifting the debate toward the relation between the data distribution and human perception. However, this view typically treats such non-robust features as fixed properties of the data rather than as quantities whose effective role may depend on network encoding strategies. Consequently, neither perspective fully explains how representational constraints interact with data semantics.

In this paper, we bridge this divide from a mechanistic-interpretability perspective, showing how adversarial vulnerability can arise through the interaction between architectural constraints and data semantics (*i.e.* human interpretable properties of the data). We argue that AExs may emerge from interference among learned representations in NNs: adversarial attacks systematically exploit superposed latent structure – a mechanism that ordinarily increases representational capacity – to construct perturbations that manipulate model outputs. Our analysis identifies a mechanistic pathway in which input correlations constrain feature arrangements, those arrangements determine interference

patterns, and the resulting interference patterns govern attack characteristics and transferability. This framework therefore supports predictions about which perturbations succeed and why they transfer between models.

Our account of AExs draws on recent theories of how networks encode and process information, in particular the linear representation hypothesis (LRH) (Park et al., 2024) and the theory of superposition (Elhage et al., 2022). The LRH posits that *input features*—fundamental abstractions of data—are represented as linear directions in a network’s representation space. Superposition further hypothesises that NNs can represent substantially more such features than they have neurons, efficiently packing multiple features into shared dimensions while thereby introducing interference. Because this interference implies that perturbing one feature may affect others in non-obvious ways, we investigate whether it creates exploitable adversarial vulnerabilities and what this mechanism reveals about adversarial phenomena. Our primary contributions are:

- We present a framework that reconciles architectural constraints with feature properties in adversarial vulnerability. Using simple models, we show that data properties induce specific superposition geometries that directly determine realised adversarial perturbations, and that interference patterns illuminate both attack transferability and class-specific susceptibility.
- We demonstrate that these findings persist in a ViT trained on CIFAR-10 with an engineered bottleneck layer that induces superposition.
- We show that superposition is sufficient, but not necessary, for adversarial vulnerability; in doing so, we identify algorithmic brittleness as a distinct vulnerability mechanism.
- We demonstrate that mechanistic understanding of learned representations enables informed adversarial attack construction, thereby motivating semantically informed defences.

2 BACKGROUND

We first introduce the two representational tools on which our analysis relies: the LRH and superposition. Formally, let $\mathbf{x} \in \mathcal{X}$ denote the input to a NN, and let $\mathbf{h}^{(l)} \in \mathbb{R}^{d_l}$ denote the activation vector at layer l .

Linear representation hypothesis (LRH). The LRH posits that NNs encode many variables relevant to their computation, including semantic properties of their inputs, as linear directions in activation space, thereby furnishing abstractions through which network behaviour can be analysed (Park et al., 2024; Guerner et al., 2023). This bias toward linear feature representation is hypothesised to arise because linear separability makes features comparatively easy for networks to recognise and manipulate, while dot products with subsequent layer weights provide an efficient mechanism for processing directional features. A growing body of empirical work supports this view (Gurnee et al., 2023; Park et al., 2024). Let $\mathcal{C} = \{c_1, \dots, c_M\}$ denote a set of M semantically meaningful latent features (e.g. concepts such as “presence of shape,” or “indoor vs. outdoor”). Formally:

Definition 1 (Linear Representation Hypothesis (LRH)) *A neural network layer with activations $\mathbf{h}^{(l)} \in \mathbb{R}^{d_l}$ satisfies the LRH if it represents the latent features $\mathcal{C} = \{c_1, \dots, c_M\}$ as linear directions $\{\mathbf{v}_j\}_{j=1}^M \subset \mathbb{R}^{d_l}$ such that:*

$$\mathbf{h}^{(l)}(\mathbf{x}) \approx \sum_{j=1}^M a_j(\mathbf{x}) \mathbf{v}_j$$

where $a_j(\mathbf{x}) \in \mathbb{R}$ represents the activation magnitude of feature c_j , and $\mathbf{v}_j$ is the corresponding linear direction. The features are **linearly accessible** (Costa et al., 2025): inputs $\mathbf{x}_0, \mathbf{x}_1 \in \mathcal{X}$ that differ mainly in the value of feature c_j while holding other features approximately constant (i.e., $a_j(\mathbf{x}_1) - a_j(\mathbf{x}_0) = k$ and $|a_i(\mathbf{x}_1) - a_i(\mathbf{x}_0)| < \lambda$ for all $i \neq j$ and some small $\lambda > 0$) satisfy:

$$\mathbf{h}^{(l)}(\mathbf{x}_1) - \mathbf{h}^{(l)}(\mathbf{x}_0) \approx k \mathbf{v}_j$$

where k reflects the change in c_j .

Superposition and sparsity. NNs can represent substantially more latent features than there are available dimensions in an activation space: $M > d_l$. For instance, LLMs can refer to vastly more place names than they have residual-stream dimensions. Superposition provides one framework for analysing this phenomenon, proposing that networks employ an overcomplete, non-orthogonal

collection of feature directions $\{\mathbf{v}_j\}_{j=1}^M$. This account exploits the fact that $2^{\Theta(d\epsilon^2)}$ almost orthogonal vectors ($< \epsilon$ cosine similarity) can be embedded in d -dimensional space (Tkocz, 2012), together with the observation that sparse vectors can often be accurately recovered after projection into lower-dimensional spaces (Elhage et al., 2022; Bereska & Gavves, 2024; Sawmya et al., 2025). This compression creates two related interpretability challenges. First, *polysemanticity* (Scherlis et al., 2022; Lecomte et al., 2023) arises when individual neurons contribute to multiple distinct features, so that a neuron’s activation no longer corresponds to a single interpretable concept. Second, non-orthogonal feature directions induce *interference* between features: activating one feature partially activates others. Networks may partially mitigate these issues through non-linear operations (e.g. ReLU, softmax) that disambiguate superposed features (Gurnee et al., 2023).

Definition 2 (Superposition Hypothesis) *A network layer represents features in superposition if:*

1. *The number of latent features exceeds layer dimensionality: $M > d_l$*
2. *There exists non-orthogonal feature directions: $\exists i, j$ with $i \neq j$ such that $\mathbf{v}_i \cdot \mathbf{v}_j \neq 0$*
3. *Latent feature activations are sparse: $\mathbb{E}_{\mathbf{x}}[\|\mathbf{a}(\mathbf{x})\|_0] \ll M$, where $\mathbf{a}(\mathbf{x}) = [a_1(\mathbf{x}), \dots, a_M(\mathbf{x})]^T$*

The network trades representational capacity against feature interference by packing more features than dimensions.

Adversarial attacks. We next define adversarial perturbations and attacks following (Szegedy et al., 2014; Costa et al., 2024):

Definition 3 (Adversarial attack) *Let $\mathbb{B}_p(\mathbf{x}, \epsilon)$ denote the ϵ -ball around $\mathbf{x} \in \mathcal{X}$ in the ℓ_p sense, so that, for example, $\mathbb{B}_\infty$ is the ℓ_∞ -ball. Any small perturbation δ that p -perturbs $\mathbf{x}$ to produce an **adversarial input** $\mathbf{x}' \in \mathbb{B}_p(\mathbf{x}, \epsilon)$ such that $f(\mathbf{x}') \neq f(\mathbf{x})$ in the untargeted case, or $f(\mathbf{x}') = y_{\text{target}}$ in the targeted case, constitutes an **adversarial attack**. ℓ_∞ - or ℓ_2 -norm attacks are generated via projected gradient descent (PGD), as detailed in Sec. A.*

Definition 4 (Adversarial robustness) *A classifier $g_w(\cdot)$ with parameters w is **robust** at $\mathbf{x}$ with radius $\epsilon > 0$ if $g_w(\mathbf{x}') = g_w(\mathbf{x}) \forall \mathbf{x}' \in \mathbb{B}(\mathbf{x}, \epsilon)$.*

The central question of this paper is therefore not merely whether such perturbations exist, but whether their structure can be mechanistically predicted from the geometry of learned feature representations.

3 SUPERPOSITION GEOMETRY CAN DETERMINE ADVERSARIAL ATTACKS

We begin by examining whether, and through what mechanism, adversarial attacks exploit interference arising from the superposition of latent features. More specifically, our analysis is organised around three questions:

1. *Do adversarial perturbations exploit the interference between superposed features?*
2. *Do correlations in the input shape the geometric arrangement of superposed latent features?*
3. *Can shared latent geometry explain why attacks transfer between independently trained models?*

Experimental setup. To investigate these questions, we construct a synthetic task that instantiates established LRH and superposition phenomena (Park et al., 2024; Elhage et al., 2022) within a precisely controllable experimental setting. The task is designed to (1) provide an intuitive classification problem for studying AExs; (2) represent class concepts as linear directions, in accordance with the LRH; (3) induce controlled superposition among these latent features; and (4) preserve a priori knowledge of the mapping from inputs to superposed features, thereby enabling testable predictions about adversarial mechanisms.

Concretely, we partition each input $\mathbf{x} \in \mathbb{R}^d$ into k equally sized groups, $\mathbf{x} = [\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(k)}]$, where $\mathbf{x}^{(j)}$ denotes the p input features associated with class j . The classification rule selects the group with the largest sum: $y = \arg\max_{j \in \{1, \dots, k\}} \sum_{i=1}^p x_i^{(j)}$. Thus, each $x_i^{(j)}$ may be interpreted as evidence for class j , with the p features corresponding to distinct class attributes. Following Elhage et al. (2022), we sample $x_i^{(j)} \sim \text{Uniform}(0, 1)$ with sparsity S (the probability of setting the feature to zero).

A two-layer network compresses the input into m dimensions through a bottleneck encoder, $\mathbf{h} = \text{ReLU}(\mathbf{W}_e \mathbf{x} + \mathbf{b}_e) \in \mathbb{R}^m$ (where $m < k < d$), and reconstructs class logits via a decoder, $\mathbf{z} =$

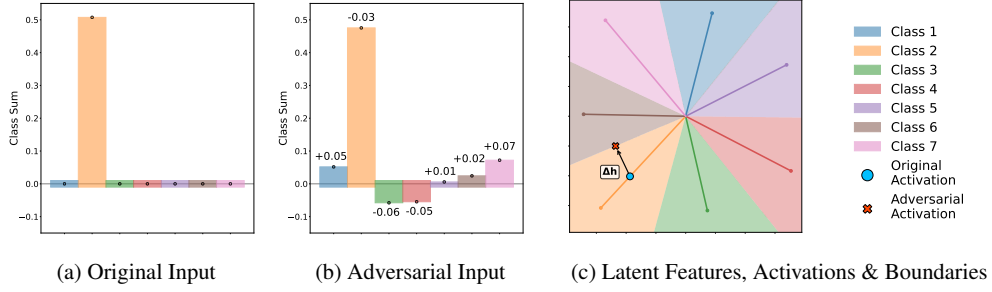


Figure 1: **An adversarial attack exploiting superposition geometry** ($k = 7$, $m = 2$). (a) The original sample. (b) The adversarially perturbed sample, whose ground truth is unchanged despite misclassification. The sign and magnitude of the input perturbation are determined by the configuration of latent representations. (c) The original and adversarial samples in activation space; arrows denote the column vectors of $\mathbf{W}_e$, corresponding to latent representations of input features.

$\mathbf{W}_d \mathbf{h} + \mathbf{b}_d \in \mathbb{R}^k$. Our primary configuration uses cross entropy (CE) loss without ReLUs or biases, a deliberately simple setting in which the softmax limits interference by exponentially suppressing contributions from interfering latent features. To verify that our conclusions are not an artifact of this particular formulation, we additionally evaluate a second configuration using mean squared error (MSE) loss with ReLUs and biases. In both cases, the representational problem is to compress information from $d = k \times p$ input dimensions into $m < k$ latent dimensions while preserving enough information to identify the group with the largest sum.

Although non-synthetic tasks contain features corresponding to nuanced concepts that must themselves be discovered and interpreted, this construction supplies ground-truth class labels for the latent features. We can therefore compute the true class after any perturbation simply by recalculating group sums, which permits us to define an AEx as satisfying the following two conditions:

1. The model’s prediction under perturbation changes (*e.g.* from class A to B)
2. The true class remains unchanged (*e.g.* still class A)

This criterion separates genuine AExs from perturbations that genuinely alter the ground truth (*i.e.* perturbations that change which group has the highest sum). We construct attacks with ℓ_∞ - and ℓ_2 -norm PGD.

Evaluation metrics. We treat the columns of the encoder weight matrix $\mathbf{W}_e$ as latent representations $\mathbf{v}_i$ for the corresponding input features x_i . Because an input feature $x_i^{(j)}$ contributes exclusively to class j , the associated $\mathbf{v}_i$ (for $i \in \text{class } j$) align with that class. This direct correspondence between input features and class-aligned latent directions allows us to trace, with unusually high precision, how perturbations to inputs of a given class, $\mathbf{x}^{(j)}$, induce changes in latent space. For successful PGD-generated AExs $\mathbf{x}_{adv} = \mathbf{x}_{orig} + \delta$, we measure:

1. **Input perturbation profile (IPP):** The sign and magnitude of each δ_i , describing how different input features are perturbed in an attack.
2. **Latent attack alignment:** The similarity $\Delta \mathbf{h} \cdot \mathbf{v}_c$ between the *latent attack vector* $\Delta \mathbf{h} = \mathbf{h}_{adv} - \mathbf{h}_{orig}$ and class c ’s representation.
3. **Attack transferability:** Success rate of attacks generated on one model when applied to another.
4. **Robust accuracy:** The fraction of examples that remain correctly classified after perturbation δ with $\|\delta\|_p \leq \epsilon$ to a sample of inputs.

3.1 RESULTS

Do attacks exploit interference between latent features in superposition? An attack must move a sample across a decision boundary in order to change its predicted class; the central question is what determines the input perturbations capable of producing that movement. Two contrasting intuitions are plausible: the *feature intuition* suggests that an attack should simply increase input features associated with the target class, whereas the *bug intuition* suggests that adversarial perturbations alter inputs in essentially unpredictable ways, so that knowledge of feature representations provides little explanatory leverage.

Fig. 1 presents a representative AEx in this setting: an input from Class 2 (orange), together with its perturbed value and IPP, is misclassified as Class 6 (brown). Although the perturbations initially appear arbitrary, they do not merely increase target-class features, contradicting the feature intuition; nevertheless, they are not random. Rather, they instantiate a precise pattern in which δ systematically corresponds to the configuration of latent features. This correspondence motivates a third intuition: *attacks are mediated by latent geometry*.

Quantitatively, the IPP exhibits a strong correspondence with latent attack alignment. The sign and magnitude of perturbations $\delta^{(j)}$ are strongly related to the alignment between the relevant class representation $\mathbf{v}^{(j)}$ and the latent attack vector $\Delta\mathbf{h}$: positively aligned representations ($\mathbf{v}^{(j)} \cdot \Delta\mathbf{h} > 0$) undergo proportional amplification of their input features, whereas negatively aligned representations are attenuated. This is precisely the structure visualised in Fig. 1(b), where the input perturbations required to cross the decision boundary are determined by the dot products shown in Fig. 1(c).

To assess whether PGD attacks specifically exploit superposition geometry, we compare PGD-generated attacks with theoretically optimal perturbations that, as derived in Sec. 3.2, explicitly leverage superposition. For each configuration (k, m) , we train 5 models with distinct random seeds, generate 1000 PGD attacks per model, and compute the cosine similarity between each successful perturbation and its corresponding optimum. As a random baseline, we generate perturbations with matching ℓ_2 norm and compute their similarity to the same theoretical optimum. Tab. 1 shows near-perfect alignment between PGD and optimal attacks across dimensionalities. To ensure that the comparison concerns substantive perturbations, we filter out samples whose ℓ_2 norm is less than ϵ . One-sample t-tests comparing observed similarities against the random baseline yield $p < 10^{-10}$ in every configuration, demonstrating that PGD attacks systematically match theoretical predictions rather than arising by chance. Because the optimal attacks exploit interference, and because PGD attacks align almost perfectly with these optima, we conclude that PGD exploits superposition interference.

Finding: Adversarial attacks systematically exploit interference between superposed features. Successful PGD attacks are predictable given the specific superposition geometry, rather than random.

Do input correlations determine latent feature geometry? We next ask whether correlations in the input distribution determine the geometric arrangement of latent features. By systematically manipulating correlation structure during training, we test how such correlations restrict the degrees of freedom available to learned representations. We introduce three correlation patterns that impose progressively stronger constraints. In the *uncorrelated* condition, features are sampled independently, as above. Under *paired* correlations, each feature in class i is coupled with a feature in class j , so that activation of one entails activation of its partner. Under *global* correlations, each sample receives a random phase that determines a cyclic activation pattern: classes adjacent in index have similar activation probabilities and therefore co-activate frequently, whereas more distant classes rarely co-activate.

Fig. 2 illustrates how these correlations shape the arrangement of $\mathbf{v}^{(j)}$. We quantify geometric similarity by comparing the pairwise cosine-similarity matrices over all feature pairs in each model and then measuring the correlation between these matrices across random seeds, with larger values indicating more similar geometric arrangements. For each correlation condition, we evaluate 25 seed pairs. In the $m = 2, k = 7$ setting, geo-

Table 1: Alignment between PGD-discovered attacks and theoretically optimal perturbations across configurations. Entries report mean $\pm$ std over 1000 attacks per condition; all p -values are below 10^{-10} .

k	m	Cosine Sim. (PGD vs Theory)	Cosine Sim. (Random Baseline)
6	2	0.97 ± 0.02	0.00 ± 0.02
30	10	0.96 ± 0.00	0.00 ± 0.01
90	30	0.92 ± 0.00	0.00 ± 0.00

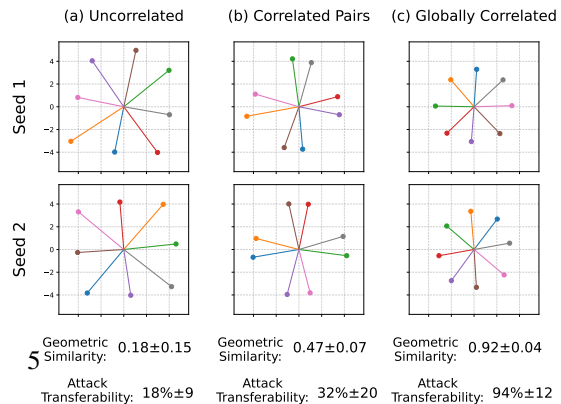


Figure 2: Stronger input correlations induce more

metric similarity is 0.23 ± 0.14 for uncorrelated data, 0.47 ± 0.32 for paired correlations, and 0.88 ± 0.06 for global correlations (mean $\pm$ std). The resulting monotonic pattern indicates that uncorrelated data permits highly variable geometries across seeds, paired correlations partially constrain the arrangement, and global correlations impose near-identical geometries. Statistical significance is assessed with pairwise two-sample t-tests between all three modes with Bonferroni correction ($p < 10^{-3}$ for all comparisons, $\alpha = 0.05$ corrected for 3 comparisons). Results across additional (k, m) configurations appear in Sec. C.3.

The operative mechanism is *interference avoidance*: features that frequently co-activate are arranged so as to minimise mutual interference. Stronger correlations correspondingly impose stronger geometric constraints: uncorrelated data permits any arrangement that separates classes, paired correlations fix the relative positions of coupled features, and global correlations determine a unique geometry up to rotation.

Finding: Input correlations constrain feature geometry. Stronger correlations reduce geometric degrees of freedom, forcing different initialisations to converge to similar arrangements.

Does shared geometry explain attack transferability? Having established that input correlations induce consistent latent geometries across model initialisations, we next test whether the associated similarity of interference patterns explains attack transferability. If attacks are specialised to particular interference patterns, they should transfer successfully primarily between models whose geometric arrangements are similar.

Using the same seed pairs as above, we generate adversarial attacks on source models and evaluate their success when applied to target models within each correlation condition, yielding 25 transfer measurements per condition. All models achieve $> 95\%$ clean accuracy, and transfer rates correlate strongly with geometric similarity. For the $k = 7, m = 2$ configuration (Fig. 2), globally correlated data yields $94\% \pm 12\%$ attack transfer (mean $\pm$ std), uncorrelated data yields only $18\% \pm 9\%$ transfer, and paired correlations produce the intermediate value $32\% \pm 20\%$.

Each perturbation component amplifies or suppresses features so as to create constructive interference that moves representations across decision boundaries. When the identical perturbation is applied to a model with different feature arrangements, components that previously produced constructive interference may instead cancel or interfere destructively, causing the attack to fail.

Finding: Attack transferability is governed by shared interference patterns.

Does reducing superposition suppress these attacks? If superposition creates vulnerability through interference, then reducing or eliminating superposition should correspondingly suppress adversarial vulnerability. We test this prediction through three interference-limiting experiments. First, we set $m = k$, allowing the network to represent each class direction $\mathbf{v}^{(j)}$ orthogonally. In this configuration, any perturbation that changes the model’s prediction also changes the ground-truth class: moving a sample from class A to class B requires making the sum of class B features exceed the sum of class A features, thereby genuinely transforming it into a class B sample. We find zero successful adversarial examples across 1000 attempts for every tested ϵ . Second, we isolate the effect of superposition by fixing m while varying k . Robust accuracy decreases monotonically with superposition pressure k/m (App. Tab. 5), showing that vulnerability scales with the degree of interference. Third, we test whether attacks preferentially allocate their budget to superposed features

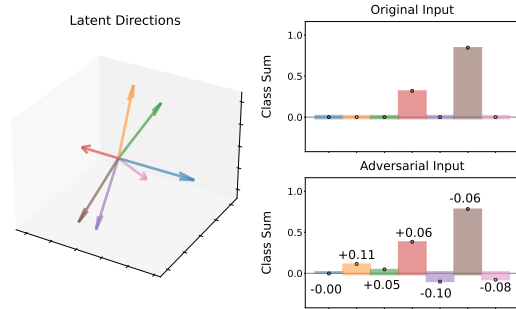


Figure 3: An adversarial attack from class 5 to class 3 leaves unperturbed the input features of a class represented orthogonally (class 1).

by constraining one class’s latent vector $\mathbf{v}^{(\perp)}$ to remain orthogonal to all others during training. When attacks are generated between classes that remain in superposition (e.g. from class a to b where $\mathbf{v}_a \not\perp \mathbf{v}_b$), the inputs associated with the orthogonal class remain unperturbed (Fig. 3).

Finding: Adversarial attacks use their budget to exploit those features in superposition.

3.2 FORMAL ANALYSIS

We now formalise the mechanisms that underlie the preceding empirical observations. In the linear setting considered here, exact relationships between superposition interference and adversarial vulnerability can be derived, thereby extending the analysis beyond observational correlation. We use the notation from our experimental setup: $\mathbf{W}_e \in \mathbb{R}^{m \times d}$ induces superposition through columns $\mathbf{v}_i$, and the decoder corresponds to the transposed encoder, as empirically observed. The logit for class j is $z_j = \mathbf{v}_j^\top \mathbf{h}$.

Proposition 1. *The optimal input perturbations δ that maximise movement from class j to class k under constraint $\|\delta\|_2 = \epsilon$ satisfy $\delta \propto \mathbf{W}_e^\top \mathbf{n}$, where $\mathbf{n} = (\mathbf{v}_k - \mathbf{v}_j)$ is the normal to the decision boundary between classes.*

Proof Sketch. To move from class j to k , we maximise the logit margin $z_k - z_j = \mathbf{n}^\top (\mathbf{h} + \Delta \mathbf{h})$, where $\mathbf{n} = (\mathbf{v}_k - \mathbf{v}_j)$ and $\Delta \mathbf{h} = \mathbf{W}_e \delta$. This becomes $\max_{\|\delta\|_2 = \epsilon} \delta^\top \mathbf{W}_e^\top \mathbf{n}$. By Cauchy-Schwarz, this is maximised when $\delta \propto \mathbf{W}_e^\top \mathbf{n}$. Full proof in Sec. B. $\square$

Corollary 1 (Interference Drives Vulnerability) *The adversarial perturbation magnitude for feature i is $|\delta_i| \propto |\mathbf{v}_i^\top (\mathbf{v}_k - \mathbf{v}_j)|$, directly proportional to the differential interference between feature i and the class representations.*

This derivation makes explicit how superposition generates adversarial vulnerability. Each input feature i is perturbed in proportion to its interference with the relevant class representations. Under superposition, non-orthogonality allows semantically unrelated features to influence the class decision; adversarial perturbations exploit precisely these induced dependencies in order to manipulate outputs.

Proposition 2. *Models with feature representations related by orthogonal transformation $\mathbf{Q}$ (where $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}$) share identical optimal attack directions in input space.*

Proof Sketch. Under transformation $\mathbf{v}'_i = \mathbf{Q} \mathbf{v}_i$, optimal perturbations remain invariant: $\delta'_i \propto (\mathbf{v}'_i)^\top (\mathbf{v}'_k - \mathbf{v}'_j) = \mathbf{v}_i^\top \mathbf{Q}^\top \mathbf{Q} (\mathbf{v}_k - \mathbf{v}_j) = \mathbf{v}_i^\top (\mathbf{v}_k - \mathbf{v}_j) = \delta_i$. Full proof in Sec. B. $\square$

Together with the empirical findings, these propositions establish a **mechanistic pathway**. Specifically, (i) input correlations constrain feature arrangements in superposition, (ii) those geometric arrangements determine interference patterns, (iii) interference patterns dictate optimal perturbations via $\delta \propto \mathbf{W}_e^\top (\mathbf{v}_k - \mathbf{v}_j)$ (Proposition 1), and (iv) shared geometric constraints produce similar interference patterns across models, thereby enabling transferability (Proposition 2):

Correlations $\xrightarrow{\text{constrain}}$ **Feature Geometry** $\xrightarrow{\text{determines}}$ **Interference Patterns** $\xrightarrow{\text{enable}}$ **Transferability**

4 ATTACKS IN REALISTIC MODELS

We next extend the analysis to more realistic architectures, focusing on a ViT (Dosovitskiy et al., 2020) trained on CIFAR-10 (Krizhevsky, 2009), with a complementary ResNet-18 (He et al., 2016) experiment reported in Sec. D.2.1. By inserting a bottleneck layer immediately before the final classification output, we again force latent class features to be represented in superposition. We then examine (i) how the degree of superposition affects attack transferability and robustness, (ii) whether a consistent arrangement of superposed representations emerges across initialisations, and (iii) whether the resulting attacks exploit feature interference.

Setup. We train a ViT architecture with 6 layers, patch size 4, and residual stream dimension $d = 512$ on CIFAR-10 across five random seeds. After training, we replace the original classification head with a bottleneck module comprising a linear encoder that compresses pre-classification activations into an m -dimensional space, followed by a linear decoder that expands back to the $k = 10$ CIFAR-10 classes. This bottleneck, trained with CE loss and no ReLU in order to mirror the setup in Sec. 3,

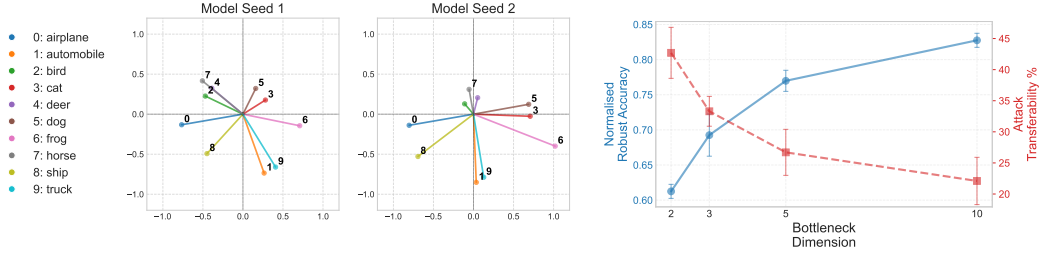


Figure 4: Left: CIFAR-10 class representation structure remains similar across models trained with different seeds. Right: attack transferability and robust accuracy as bottleneck dimension increases.

forces the k class representations into m -dimensional superposition while the weights of the base ViT remain frozen. We vary the bottleneck dimension $m \in \{2, 3, 5, 10\}$ to control the severity of superpositional compression. AExs are generated for each bottlenecked model using both ℓ_∞ - and ℓ_2 -norm PGD. We use a small step size and many iterations, terminating when an attack successfully changes the model’s prediction; full details are provided in Sec. D.1. Robust accuracy is normalised by clean accuracy to account for performance differences induced by varying bottleneck capacity, and attack transferability is measured as defined in Sec. 3.

Results. The mean clean accuracy of the base models is 81%. As expected, clean accuracy in the bottlenecked models decreases as compression intensifies, yielding 54%, 67%, 76%, and 81% for $m = \{2, 3, 5, 10\}$, respectively.

We make three observations that closely parallel the toy-model results, providing evidence that the proposed mechanism transfers to this more realistic setting. First, Fig. 4 (right) shows that as the bottleneck dimension m decreases (*i.e.* as superposition pressure increases), normalised robust accuracy decreases while attack transferability across model initialisations increases. The robust-accuracy decrease is plausibly attributable to denser packing of class representations, which makes small input perturbations more capable of producing widespread interference among compressed features. The increase in transferability follows the mechanism identified in Sec. 3: stronger superposition reduces the degrees of freedom available to feature geometry, leaving fewer viable arrangements for class representations. As a result, different model initialisations converge toward more similar representational geometries, yielding more similar interference patterns and therefore more transferable attacks. Results across perturbation magnitudes are provided in Sec. D.1.1.

Second, Fig. 4 (left) shows that the geometric arrangement of the m -dimensional class representations in the bottleneck exhibits consistent ordering and clustering across models. This consistency, analogous to the behaviour in Sec. 3, suggests that intrinsic correlations in the data guide the formation of superposed geometry. Notably, the resulting groupings often reflect semantic similarity between classes, such as clustering of ‘cat’ and ‘dog’ representations. Third, when AExs generated for a bottlenecked model are passed through the corresponding base ViT, they induce similar relative logit changes (*i.e.* $\text{logit}_{\text{base}}(\mathbf{x}_{\text{adv}}) - \text{logit}_{\text{base}}(\mathbf{x}_{\text{orig}})$). Thus, both the bottlenecked and base models respond coherently to the same perturbations, indicating that the perturbations are not merely artifacts of the bottleneck, although without the increased compression they do not suffice to cause misclassification.

5 ADVERSARIAL VULNERABILITY VIA ALGORITHMIC BRITTLINESS

The preceding sections argued that feature superposition can *suffice* for adversarial vulnerability. We now ask whether it is also *necessary*, studying a setting in which features are represented orthogonally rather than in superposition, yet remain vulnerable to attack. This case supports the paper’s broader claim that analysing feature representations can be useful for robustness even beyond superpositional mechanisms. We organise the section around three questions: *how do such attacks work mechanistically, can this understanding be used to construct attacks directly, without gradients, and does naive robustness certification alter the underlying failure mode or merely increase required attack budgets?*

Setup. We train a linear encoder + MLP on one-hot integers (a, b) to predict $(a + b) \bmod P$. As detailed in Sec. E and Nanda et al. (2023), the model discovers a trigonometric algorithm for solving

this task. As part of this learned algorithm¹, the encoder projects inputs onto non-basis-aligned linear representations encoding $\sin(\omega_k a)$ and $\cos(\omega_k a)$ at key frequencies $\omega_k = 2\pi k/P$. The MLP then composes $\cos((a+b)\omega_k)$ via sum-angle identities, achieving 100% accuracy. Crucially, these features are represented *orthogonally* (i.e. not in superposition). We generate AExs by relaxing the one-hot constraint and perturbing only input a with PGD, thereby moving inputs off the training manifold.

Attacks exploit the learned key frequencies. Fourier analysis of successful perturbations δ (App. Fig. 6) reveals that their dominant frequencies coincide precisely with the learned algorithm’s key frequencies, ω_k . Moreover, 0% of attacks transfer to models with different ω_k , such as models trained from different seeds. The frequency-aligned perturbations induce misclassification by altering the sinusoidal values read from the encoder features: the perturbation δ transforms the one-hot vector $\mathbf{x}_a$ into a dense vector $\mathbf{x}'_a$, and the newly non-zero inputs introduce contributions that corrupt the sinusoidal values subsequently used by the MLP. Because the model’s algorithm relies on constructive interference of each ω_k at the logits, disrupting a single ω_k is insufficient; successful attacks must instead identify δ that substantially alters multiple frequency components. We provide a more detailed intuition for this mechanism in Sec. E.1.

Analytic construction of attacks. Using this mechanism, we can construct gradient-free, *informed* perturbations consisting of composite sinusoidal waves at matching ω_k . On $0+0 \bmod P$, these attacks succeed with $\|\delta\|_2=0.22$, close to PGD (0.17, $\approx 1.3\times$ the PGD budget) and substantially below naive alternatives (single-index 0.97, ℓ_∞ noise 2.04, uniform 6.20). This provides empirical evidence that the constructed attacks target the correct mechanism and illustrates that a representation-aware account of vulnerability can yield concrete attack recipes rather than only post-hoc explanations.

Do defences mitigate this algorithmic vulnerability? We apply certified training against ℓ_∞ perturbations (Müller et al., 2023). For fixed attack budgets, *robust* accuracy improves from pre-robustified $\rightarrow$ post-robustified as follows: 0% $\rightarrow$ 85% at $\epsilon_\infty = 5e-3$, 0% $\rightarrow$ 96% at $2e-3$, and 76% $\rightarrow$ 100% at $1e-3$. Nevertheless, when attacked with PGD, the robust model continues to fail through the same frequency-based brittleness, albeit at larger perturbation norms. Certification therefore increases the required budgets without eliminating the specific fragility, motivating defences that target identified semantics, here frequency manipulation, rather than relying exclusively on broad norm bounds.

Relation to the paper’s thesis. This section contributes two points to the paper’s broader narrative. First, superposition is *sufficient but not necessary*: even orthogonal representations can remain adversarially vulnerable. Second, analysing AExs through the lens of *architecture–semantics interaction* is practically actionable, because it can guide both attack construction through representation-informed perturbations and defence development.

Although less central in this setting, a shared dependence on sparsity assumptions remains: upstream violations of sparsity (e.g. noise or superposed features) can deliver effectively ‘pre-perturbed’ inputs to downstream orthogonal circuits, thereby activating the same frequency-based failure mode.

6 RELATED WORK

Superposition & latent representations. Research on superposition has identified several mechanisms through which feature interference arises: Nanda (2023) distinguish *representational* from *computational* superposition, while Gurnee et al. (2023) differentiate *alternating* and *simultaneous* interference. Data correlations have likewise been shown to shape representational arrangements: correlated feature pairs tend to become orthogonal (Elhage et al., 2022), correlations can drive the formation of superposition (Chan, 2024), and semantic clustering can emerge from correlation structure (Prieto et al., 2025a). More broadly, representation geometry is influenced by spectral bias (Rahaman et al., 2019), neural collapse (Kothapalli, 2023), and optimisation objectives (Casper, 2023). Sparse autoencoders (SAEs) have also been applied to vision models in order to disentangle features represented in superposition (Lim et al., 2024; Joseph et al., 2025).

¹We refer to the specific computation executed by the network to solve the task as the *model’s learned algorithm*. As this algorithm depends on extracting precise values from learned feature representations (frequencies), it remains *brittle* in the face of small perturbations to the bases. We term this the *algorithmic brittleness*.

Adversarial vulnerability. The adversarial-robustness literature is often organised around an attack–defence dichotomy. Attacks (Szegedy et al., 2014) use imperceptible perturbations to maximise model error, with canonical examples including FGSM (Goodfellow et al., 2014) and PGD (Madry et al., 2018). Adversarial robustness, conversely, concerns the maintenance of performance under such attacks. Defences include adversarial training (Madry et al., 2018) and certified training (Wong & Kolter, 2018; Palma et al., 2024), which provide guarantees against perturbations within specified input bounds.

The role of input & latent features on adversarial robustness. Prior work attributes vulnerability either to predictive but brittle ‘non-robust’ input features (Ilyas et al., 2019; Goh, 2019) or to learned shortcuts (Li et al., 2023), and perturbation characteristics have been shown to vary across datasets (Maiya et al., 2021). Elhage et al. (2022) suggest that superposition may be linked to AExs, although this connection is debated by Casper (2023). A tension between representational efficiency and adversarial robustness has also been observed (Barsbey et al., 2025). Zhang et al. (2021) show that adversarial perturbations push latent representations across decision boundaries, while Ganeshan et al. (2019) find that PGD targets final layers and propose feature-based attacks. Finally, transferability is constrained by representation discrepancies across models (Li et al., 2023; Wang et al., 2024), and decorrelating features reduces transfer (Wiedeman & Wang, 2022), aligning with our findings.

7 DISCUSSION & CONCLUDING REMARKS

We have shown that adversarial attacks can exploit interference patterns induced by the geometry of superposed features in NNs. Across toy models and ViTs, our experiments indicate that data properties, especially correlations and sparsity, induce distinct superposition geometries that in turn give rise to predictable adversarial vulnerabilities. These geometries, together with the interference patterns they generate, support predictions about attack transferability and class-specific susceptibility. The resulting perspective reframes adversarial vulnerability as a possible inherent consequence of efficient representational compression, rather than solely as a failure of optimisation, data quality, or decision-boundary regularity.

Limitations & future work. Our insights derive primarily from simplified models. Although the analyses provide an initial mechanistic lens on attacks in larger models, the extensive adversarial-robustness literature suggests that precise vulnerability mechanisms likely vary with both attack type and model class. For example, constrained attacks such as universal perturbations, jailbreaks, or patch attacks may be unable to activate the particular feature combinations needed to exploit superposition interference.

Future work should quantify the ‘adversarial cost’ of superposition, clarify its relationship to accuracy–robustness trade-offs, and characterise how robust training reshapes superposition geometry. Our current evidence comes from simplified settings in which class features are placed in engineered superposition, whereas real networks contain interference among unknown, unlabelled features distributed across multiple layers. We plan to use SAEs (Bricken et al., 2023) to address these challenges in large-scale networks, with preliminary results in Sec. F indicating that this is a promising direction.

ETHICS STATEMENT

Given its investigative scope and generality, our mechanistic-interpretability framework is not expected to produce specific negative impacts or immediate societal consequences. The experiments use only publicly available datasets (CIFAR-10) and synthetic toy models, and they involve neither human subjects nor personally identifiable information. The gradient-based attack methods used in the paper are already well established in the literature. Accordingly, our work does not introduce novel attack capabilities; rather, it provides theoretical and mechanistic understanding of existing phenomena. We have no conflicts of interest to declare. By clarifying the relationship between superposition and adversarial vulnerability, we aim to inform future research that balances model capability with robustness and thereby contributes to more trustworthy AI systems suitable for deployment in critical applications.

REPRODUCIBILITY STATEMENT

We include source code for all experiments in the supplementary materials. The toy-model experiments (Sec. 3) provide complete architectural specifications, training procedures, evaluation metrics, and data-generation processes. For the ViT experiments (Sec. 4), architecture and training procedures are described in the main text, with further details in Sec. D.1. The modular-arithmetic experiments (Sec. 5) specify the network architecture, data encoding, and learned trigonometric algorithm, with additional details in Sec. E.2. Experiments were conducted on an AMD Ryzen Threadripper 3970X 32-Core with 256GB RAM and RTX 3090. Additional ResNet-18 and SAE experiments are documented in Appendices Sec. D.2.1 and Sec. F, respectively.

LLM USAGE STATEMENT

This work used LLMs to assist with prior-literature search, automate LaTeX table generation from CSV files (with all numerical values subsequently verified), and suggest alternative phrasings to improve manuscript clarity. All substantive research contributions and conclusions remain the original work of the authors.

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APPENDIX

A GENERATION OF ADVERSARIAL EXAMPLES

We use PGD, an iterative attack-generation procedure (Madry et al., 2018), to construct AExs. For untargated attacks, the procedure maximises the classifier’s loss $\mathcal{L}(f(\mathbf{x}'), y_{\text{true}})$; for targeted attacks, it instead minimises $\mathcal{L}(f(\mathbf{x}'), y_{\text{target}})$ according to the update rule:

$$\mathbf{x}'^{(k+1)} = \Pi_{\mathcal{S}} \left(\mathbf{x}'^{(k)} \pm \alpha \mathbf{g}_k \right) \quad (1)$$

(+ for maximisation, – for minimisation). $\mathbf{x}'^{(0)}$ is an initial perturbed input, α is step size, $\mathbf{g}_k$ is the (normalised) gradient $\nabla_{\mathbf{x}} \mathcal{L}(f(\mathbf{x}'^{(k)}), y_{\text{class}})$ (y_{class} being y_{true} or y_{target}), and $\Pi_{\mathcal{S}}$ projects onto the ϵ -ball $\mathcal{S} = \{\mathbf{x}' \mid \|\mathbf{x}' - \mathbf{x}\|_p \leq \epsilon\}$, with input domain clipping. We use the ℓ_p norms:

1. ℓ_{∞} : Constraint $\|\delta\|_{\ell_{\infty}} = \max_i |\delta_i| \leq \epsilon$. $\mathbf{g}_k$ is $\text{sign}(\nabla_{\mathbf{x}} \mathcal{L}(\cdot))$. $\Pi_{\mathcal{S}}$ clips each $\mathbf{x}'_i$ to $[\mathbf{x}_i - \epsilon, \mathbf{x}_i + \epsilon]$.
2. ℓ_2 : Constraint $\|\delta\|_{\ell_2} = \sqrt{\sum_i \delta_i^2} \leq \epsilon$. $\mathbf{g}_k$ is $\nabla_{\mathbf{x}} \mathcal{L}(\cdot) / \|\nabla_{\mathbf{x}} \mathcal{L}(\cdot)\|_{\ell_2}$. $\Pi_{\mathcal{S}}$ rescales $\delta = \mathbf{x}' - \mathbf{x}$ if $\|\delta\|_{\ell_2} > \epsilon$ via $\delta \leftarrow \epsilon \cdot \delta / \|\delta\|_{\ell_2}$.

B OMITTED PROOFS & THEORETICAL RESULTS

To clarify how adversarial attacks exploit feature representations, we prove that optimal perturbations weight each input dimension according to the extent to which its corresponding feature aligns with the path toward the decision boundary. The analysis concerns our linear models without ReLU activation functions.

Consider an input $\mathbf{x} \in \mathbb{R}^d$ encoded by $\phi(\mathbf{x}) = \mathbf{W}_e \mathbf{x}$ into a latent representation $\mathbf{h} \in \mathbb{R}^m$, where the columns of $\mathbf{W}_e \in \mathbb{R}^{m \times d}$ form an overcomplete dictionary $\{\mathbf{v}_i\}_{i=1}^d$ with $m < d$. For our argmax task, we empirically observe that the trained encoder–decoder pair uses the encoder transpose as the decoder, $\mathbf{W}_d = \mathbf{W}_e^{\top}$. Consequently, the logit for class j is $z_j = \mathbf{v}_j^{\top} \mathbf{h}$, where $\mathbf{v}_j$ denotes the j -th column of $\mathbf{W}_e$, and the predicted class is $\hat{y} = \arg \max_j z_j$. Input perturbations map into latent perturbations as $\Delta \mathbf{h} = \mathbf{W}_e \delta$, where $\delta = (\delta_1, \dots, \delta_d)^{\top}$ contains the perturbation coefficients.

The decision boundary between classes j and k is the set of latent points at which their logits are equal, namely $\mathcal{B}_{jk} = \{\mathbf{h} : z_j = z_k\}$, equivalently $\{\mathbf{h} : (\mathbf{v}_k - \mathbf{v}_j)^{\top} \mathbf{h} = 0\}$. The vector $\mathbf{n} = (\mathbf{v}_k - \mathbf{v}_j)$ is therefore normal to this pairwise decision boundary and points in the direction that maximally increases the margin $z_k - z_j$. We briefly restate the propositions before giving their proofs.

Proposition 1 (Optimal targeted attack). The optimal input perturbations δ that maximise movement from class j toward class k under constraint $\|\delta\|_2 = \epsilon$ satisfy:

$$\delta \propto \mathbf{W}_e^{\top} \mathbf{n}$$

Proof. To move a sample from class j toward class k , one must maximise the logit margin $z_k - z_j$. Under perturbation δ , the resulting margin is:

$$z'_k - z'_j = \mathbf{v}_k^{\top} (\mathbf{h} + \Delta \mathbf{h}) - \mathbf{v}_j^{\top} (\mathbf{h} + \Delta \mathbf{h}) \quad (2)$$

$$= (\mathbf{v}_k - \mathbf{v}_j)^{\top} (\mathbf{h} + \mathbf{W}_e \delta) \quad (3)$$

$$= (\mathbf{v}_k - \mathbf{v}_j)^{\top} \mathbf{h} + (\mathbf{v}_k - \mathbf{v}_j)^{\top} \mathbf{W}_e \delta \quad (4)$$

The change in margin is $(\mathbf{v}_k - \mathbf{v}_j)^{\top} \mathbf{W}_e \delta = \delta^{\top} \mathbf{W}_e^{\top} (\mathbf{v}_k - \mathbf{v}_j)$. We seek:

$$\max_{\delta} \delta^{\top} \mathbf{W}_e^{\top} (\mathbf{v}_k - \mathbf{v}_j) \quad \text{subject to} \quad \|\delta\|_2 = \epsilon$$

Let $\mathbf{g} = \mathbf{W}_e^{\top} (\mathbf{v}_k - \mathbf{v}_j)$. By the Cauchy-Schwarz inequality:

$$|\delta^{\top} \mathbf{g}| \leq \|\delta\|_2 \|\mathbf{g}\|_2 = \epsilon \|\mathbf{g}\|_2$$

Equality is achieved when δ and $\mathbf{g}$ are parallel. Given the constraint $\|\delta\|_2 = \epsilon$:

$$\delta = \frac{\epsilon}{\|\mathbf{W}_e^{\top} (\mathbf{v}_k - \mathbf{v}_j)\|_2} \mathbf{W}_e^{\top} (\mathbf{v}_k - \mathbf{v}_j)$$

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Corollary (Interference drives vulnerability). For a targeted attack from class j to class k , the adversarial perturbation magnitude for input feature i is:

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$$|\delta_i| \propto |\mathbf{v}_i^\top (\mathbf{v}_k - \mathbf{v}_j)|$$

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where $\mathbf{v}_i$ is the i -th column of $\mathbf{W}_e$. This quantity represents the differential interference between feature i and the class representations.

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This result exposes the mechanism by which superposition produces adversarial vulnerability. Each input feature i is perturbed in proportion to $\mathbf{v}_i^\top (\mathbf{v}_k - \mathbf{v}_j)$, the differential interference between that feature and the relevant class representations. Under superposition, non-orthogonality implies that even semantically unrelated features may have non-zero inner products with $(\mathbf{v}_k - \mathbf{v}_j)$, generating exploitable interference patterns. The multi-class setting further amplifies this vulnerability, because k classes induce multiple pairwise boundaries, each associated with a distinct interference pattern. Adversarial perturbations leverage these cross-feature dependencies by manipulating features through their interference with class representations. This explains the empirical vulnerability we observe: attacks succeed not by simply increasing target-class features, but by exploiting the interference web created by superposition.

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Prop. 1 characterises the optimal perturbation direction for moving from class j to class k , providing the gradient for maximising the margin $z_k - z_j$. When intervening classes exist (where $z_j < z_i < z_k$ for some class i), following this gradient might cause the model to predict i before reaching target k . Iterative methods like PGD handle this by recomputing gradients at each step—the global attack path emerges from repeated local decisions rather than a single optimisation. Furthermore, while our analysis assumes $\mathbf{W}_d = \mathbf{W}_e^\top$ based on our empirical observations, the framework extends to arbitrary decoders. In the general case, with decoder $\mathbf{W}_d \in \mathbb{R}^{k \times m}$ having rows $\mathbf{w}_j^\top$, the optimal perturbation becomes $\delta \propto \mathbf{W}_e^\top (\mathbf{w}_k - \mathbf{w}_j)$, capturing interference between encoder directions and decoder weights.

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Proposition 2 (Attack transferability). Consider encoders ϕ and ψ with basis matrices $\mathbf{W}_e \in \mathbb{R}^{m \times d}$ and $\mathbf{W}'_e \in \mathbb{R}^{m \times d}$ whose columns are related by orthogonal transformation $\mathbf{v}'_i = \mathbf{Q}\mathbf{v}_i$ (where $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}$). If both models use their encoder transpose as decoder (i.e., $\mathbf{W}_d = \mathbf{W}_e^\top$ and $\mathbf{W}'_d = (\mathbf{W}'_e)^\top$), then both models have identical optimal input perturbation vectors for any targeted attack from class j to class k .

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Proof. For model ϕ , the optimal perturbation from Prop. 1 is:

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$$\delta^\phi \propto \mathbf{W}_e^\top (\mathbf{v}_k - \mathbf{v}_j)$$

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For model ψ with transformed columns $\mathbf{v}'_i = \mathbf{Q}\mathbf{v}_i$:

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$$\delta^\psi \propto (\mathbf{W}'_e)^\top (\mathbf{v}'_k - \mathbf{v}'_j) = (\mathbf{W}'_e)^\top (\mathbf{Q}\mathbf{v}_k - \mathbf{Q}\mathbf{v}_j)$$

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Since $\mathbf{W}'_e = \mathbf{Q}\mathbf{W}_e$ (all columns are transformed), we have:

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$$\delta^\psi \propto (\mathbf{Q}\mathbf{W}_e)^\top \mathbf{Q}(\mathbf{v}_k - \mathbf{v}_j) \tag{5}$$

$$= \mathbf{W}_e^\top \mathbf{Q}^\top \mathbf{Q}(\mathbf{v}_k - \mathbf{v}_j) \tag{6}$$

$$= \mathbf{W}_e^\top (\mathbf{v}_k - \mathbf{v}_j) \tag{7}$$

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where the last equality uses $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}$. Thus δ^ϕ and δ^ψ have identical proportionality. Under the same norm constraint $\|\delta\|_2 = \epsilon$, we have $\delta^\phi = \delta^\psi$. □

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Prop. 2 explains why attacks transfer between models trained under similar regimes. When models learn feature representations that differ only by an orthogonal transformation, they instantiate essentially the same geometric structure in different orientations and therefore share identical vulnerability patterns in input space. Because the orthogonal transformation preserves all feature inner products, it also preserves the interference patterns exploited by attacks. This accounts for our empirical observation that models trained on data with the same correlations exhibit high transferability: they discover similar feature geometries up to rotation, thereby inheriting shared adversarial vulnerabilities.

C TOY MODEL EXPERIMENTS

This section provides supplementary details, extended results, and additional discussion for the toy-model experiments in the main paper. We report model accuracies across a broader parameter range than is shown in the main text, thereby clarifying how model capacity and data characteristics such as sparsity influence learning and the emergence of feature superposition. We then provide additional visual examples corresponding to Fig. 1, illustrating adversarial mechanisms under several conditions.

C.1 HYPOTHESES TESTING FRAMEWORK

We explicitly state the hypotheses associated with the three research questions in Sec. 3.

Research Q1: Do adversarial perturbations exploit superposition geometry?

- H_0 : Adversarial perturbations are random with respect to feature geometry.
- H_1 : Adversarial perturbations systematically exploit geometric relationships between superposed representations.

Research Q2: Do data correlations determine superposition geometry?

- H_0 : Input correlations have no systematic effect on learned geometries.
- H_1 : Input correlations determine geometric arrangements across model initializations.

Research Q3: Does shared geometry explain attack transferability?

- H_0 : Attack transferability is independent of geometric similarity.
- H_1 : Transferability increases with shared latent structure.

We evaluate these hypotheses through controlled experiments:

- $H_1(1)$: We measure the input perturbation profile alignment with a class’s latent representation and the latent attack vector.
- $H_1(2)$: We systematically vary input correlations and measure resulting geometries.
- $H_1(3)$: We quantify transferability rates across models with varying geometric similarity.

C.2 ACCURACY OF TOY MODEL FOR A RANGE OF PARAMETERS

The toy-model experiments in the main paper primarily use low-dimensional settings for conceptual clarity. To characterise the model’s behaviour more broadly, this subsection reports the classification accuracies achieved by the CE toy model across hidden-layer size (h), number of classes (k), number of features, and sparsity level (S), thereby indicating when the models learn to represent features in superposition. The sparsity level denotes the probability that an individual input feature $x_i^{(j)}$ is set to zero, with larger S corresponding to greater input sparsity. We report performance as a function of feature density $1 - S$, the probability that a feature is non-zero. Tab. 2 and Tab. 3 contextualise the model’s performance limits and its capacity to learn latent representations in superposition.

C.3 CORRELATIONS CONSTRAIN ARRANGEMENTS

Tab. 4 displays the results for the experiments regarding *Do Input Correlations Determine Latent Feature Geometry?* across a wider range of classes (k) and hidden dimension size (m).

C.4 HOW THE TOY MODEL MAPS TO REAL-WORLD MODELS

How does scaling networks affect adversarial vulnerability? Although empirical evidence suggests that larger models tend to exhibit somewhat greater adversarial robustness, the effect is modest. When adversarial training is used, clearer scaling trends emerge, but the resulting improvements remain largely specific to the attack type used during training rather than constituting broad robustness (Howe et al., 2025).

Table 2: Classification accuracy of the CE toy model with fixed bottleneck dimension ($m = 2$), reported across numbers of classes (k), total input features (features = $k \times 3$), and input feature densities ($1 - S$). The results show how input sparsity modulates performance degradation as more classes must be superposed within a constrained latent space.

Classes (k)	Features	Hidden (m)	Accuracy at Input Feature Density ($1 - S$)							
			1.0	0.57	0.33	0.19	0.11	0.06	0.04	0.02
3	9	2	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
4	12	2	0.67	0.71	0.82	0.92	0.98	0.99	1.00	1.00
5	15	2	0.53	0.50	0.65	0.77	0.89	0.95	0.98	0.99
6	18	2	0.40	0.43	0.51	0.66	0.82	0.93	0.97	0.99
7	21	2	0.34	0.34	0.40	0.53	0.73	0.87	0.95	0.98
8	24	2	0.30	0.30	0.33	0.40	0.63	0.82	0.93	0.97
9	27	2	0.24	0.26	0.30	0.35	0.57	0.75	0.89	0.96
10	30	2	0.22	0.24	0.26	0.31	0.50	0.72	0.87	0.95
15	45	2	0.14	0.15	0.16	0.17	0.25	0.40	0.65	0.86
20	60	2	0.10	0.10	0.11	0.13	0.15	0.24	0.44	0.76
25	75	2	0.07	0.08	0.09	0.10	0.12	0.16	0.26	0.62
30	90	2	0.06	0.07	0.07	0.07	0.09	0.11	0.21	0.45

Table 3: Classification accuracy of the CE toy model across numbers of classes (k), total input features, bottleneck dimensions (m), and input feature sparsity levels (S), with different numbers of features per class (p).

Classes (k)	Features	Hidden (m)	Accuracy at Input Feature Density ($1 - S$)							
			1.0	0.57	0.33	0.19	0.11	0.06	0.04	0.02
30	30	30	0.23	0.24	0.38	0.62	0.83	0.94	0.99	1.00
30	90	90	0.27	0.27	0.33	0.41	0.51	0.67	0.85	0.94
40	40	30	0.67	0.54	0.73	0.77	0.88	0.96	0.99	0.99
40	120	30	0.71	0.64	0.65	0.72	0.73	0.79	0.89	0.96
60	60	10	0.05	0.07	0.12	0.25	0.47	0.73	0.90	0.97
60	180	10	0.08	0.10	0.13	0.17	0.22	0.32	0.52	0.75
80	80	30	0.15	0.17	0.23	0.41	0.63	0.79	0.91	0.98
80	240	30	0.23	0.22	0.31	0.41	0.48	0.53	0.66	0.80
100	100	10	0.03	0.04	0.05	0.10	0.21	0.43	0.69	0.87
100	300	10	0.04	0.05	0.06	0.09	0.12	0.15	0.25	0.45

Table 4: Geometric similarity results across correlation types and superposition configurations.

Correlation Type	k	m	Geometric Similarity
Uncorrelated	6	2	0.18 ± 0.15
	30	10	0.17 ± 0.02
	90	30	0.17 ± 0.01
Paired	6	2	0.47 ± 0.07
	30	10	0.26 ± 0.07
	90	30	0.25 ± 0.01
Global	6	2	0.92 ± 0.04
	30	10	0.88 ± 0.01
	90	30	0.80 ± 0.01

Two competing phenomena are relevant when models scale. Larger models possess greater capacity for representing concepts, yet general tasks such as next-token prediction over internet text appear to contain a long tail of useful concepts that a larger model can exploit. Consequently, despite increased capacity, superposition appears to remain prevalent even in frontier models (Lindsey et al., 2025). Dictionary-learning evidence supports this interpretation: SAEs require increasingly large dictionaries for larger models (Gao et al., 2025), suggesting that the number of represented features scales with model size. Thus, the fundamental tension driving superposition – the need to compress many features into limited dimensions – does not vanish with scale.

Because both superposition and adversarial vulnerability persist in large-scale models, we expect the present insights to remain relevant across scales. Understanding how superposition geometry changes with scale, and whether those changes can be exploited to mitigate vulnerability, is therefore an important direction for future work.

C.4.1 SEPARATING SUPERPOSITION EFFECTS FROM CAPACITY REDUCTION

To disentangle superposition effects from capacity reduction, we control capacity by holding the bottleneck dimension fixed. Specifically, we fix $m = 2$ and vary the number of classes k , thereby isolating superposition pressure. Model accuracy, robust accuracy, and transferability are reported in Tab. 5. These controls show that increased superposition pressure (k/m), independently of model capacity, drives the adversarial vulnerabilities we observe.

Table 5: Performance metrics stratified by correlation type, perturbation budget (ϵ), and number of classes (k).

Correlation Type	Parameters		Performance (%)		
	ϵ	k	Accuracy	Robust Acc.	Attack Trans.
Uncorrelated	0.05	2	100.0	100.0	0.0
		4	100.0	94.9	21.2
		6	99.9	87.0	9.8
		8	99.8	77.7	17.5
	0.1	2	100.0	100.0	0.0
		4	100.0	88.1	9.8
		6	99.7	75.4	9.9
		8	99.6	56.3	23.5
	0.5	2	100.0	100.0	0.0
		4	100.0	99.4	0.0
		6	99.9	62.9	30.5
		8	99.5	24.9	39.0
Fully Correlated	0.05	2	100.0	100.0	0.0
		4	100.0	95.8	98.4
		6	97.3	70.6	98.4
		8	93.0	57.4	97.2
	0.1	2	100.0	100.0	0.0
		4	100.0	88.7	98.5
		6	97.7	45.6	99.7
		8	94.2	34.8	98.6
	0.5	2	100.0	100.0	0.0
		4	100.0	90.0	100.0
		6	97.2	34.6	100.0
		8	93.0	32.0	100.0

C.4.2 ADDITIONAL EXAMPLES OF AEXS IN TOY MODEL

Fig. 1 demonstrates how adversarial attacks exploit interference between latent features in superposition. Here, we provide additional visual examples (Fig. 5) to reinforce the underlying intuition. Specifically, we supplement the main text with the following cases:

- Fig. 5a shows an additional instance of the Sec. 3.1 setup ($m = 2, k = 7$) under an ℓ_2 -norm PGD attack, illustrating the IPP and latent-space manipulations that produce misclassification.
- Fig. 5b presents the analogous setup ($m = 2, k = 7$) using an ℓ_∞ -norm PGD attack on a less sparse input.
- Fig. 5c shows an example with increased bottleneck dimensionality ($m = 3, k = 7$) and a corresponding ℓ_2 -norm PGD attack, together with the feature-vector similarity matrix used to compute geometric similarity.

D CIFAR-10 EXPERIMENTS

To assess whether the principles observed in toy models extend to more complex settings, Sec. 4 introduces CIFAR-10 (Krizhevsky, 2009) experiments with an engineered bottleneck. This appendix section provides additional details about that setup and reports extended results.

D.1 ARCHITECTURE & TRAINING INFORMATION

The base ViT (Dosovitskiy et al., 2020) architecture comprises 6 transformer layers, embedding dimension (d) 512, and 8 attention heads per transformer layer. CIFAR-10 images of size 32×32 pixels are partitioned into 4×4 patches. The Multilayer Perceptron (MLP) within each transformer block has hidden dimension 512, and learned positional embeddings are used.

The bottleneck architecture consists of a linear encoder followed by a linear decoder. The encoder projects pre-classification activations from the ViT backbone into an m -dimensional latent space ($\{2, 3, 5, 10\}$), after which the decoder maps these m -dimensional representations back to the $k = 10$ dimensions corresponding to CIFAR-10 classes.

The base ViT is trained on CIFAR-10 for 250 epochs using CE loss, Adam with default PyTorch parameters, learning rate 0.001, batch size 512, dropout, and a cosine-annealing learning-rate scheduler. Training is repeated across five random seeds to account for variability. After the base ViT is trained, its weights are frozen and the bottleneck layer is trained for 30 epochs with learning rate 0.001. Preprocessing applies RandomCrop with 4-pixel padding, resizing to the target size (32×32 by default), and RandomHorizontalFlip for data augmentation.

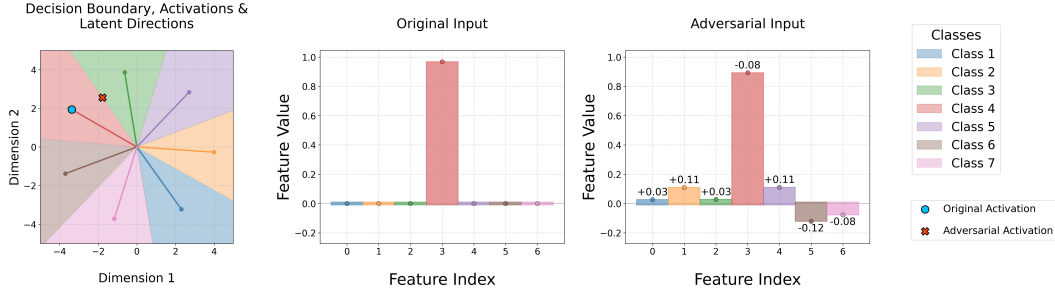
D.1.1 NORMALISED ROBUST ACCURACY ACROSS PERTURBATION MAGNITUDES

Fig. 4 (right) shows normalised robust accuracy as a function of bottleneck dimension for a single value of ϵ . This subsection extends that analysis by reporting normalised robust accuracy under PGD attacks of varying strengths. Results are provided for both ℓ_2 -norm (Tab. 7) and ℓ_∞ -norm (Tab. 6) PGD attacks, yielding a more complete view of how superposition degree interacts with attack strength to shape robustness.

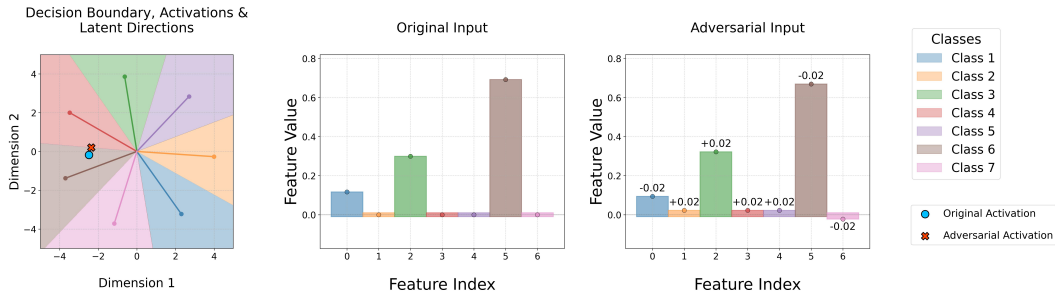
For these evaluations, AEXs are generated with PGD for 100 iterations using step size (α) 0.01. Robust accuracy is evaluated on 500 samples per configuration, and we report mean normalised robust accuracy together with the standard deviation across five random seeds.

D.1.2 ATTACK TRANSFERABILITY ACROSS PERTURBATION MAGNITUDES

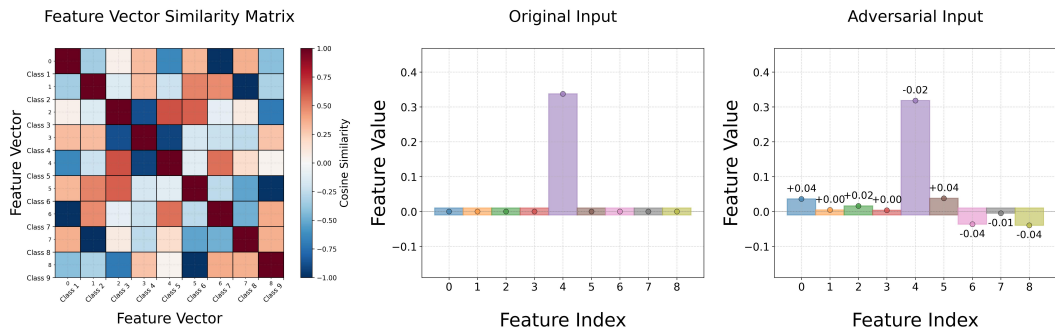
We additionally report attack transferability across perturbation magnitudes (ϵ) and bottleneck dimensions (m). Tab. 9 presents ℓ_2 -norm attack transferability, while Tab. 8 reports ℓ_∞ -norm attack transferability.



(a) An ℓ_2 -norm attack changing the classification of a class-4 input to class 3. The left panel shows original and adversarial activations in latent space, together with representation directions; the right panels show original and perturbed input-feature values.



(b) An ℓ_∞ -norm attack changing the classification of a class-6 input to class 4. The left panel shows original and adversarial activations in latent space relative to class latent directions; the right panels show original and perturbed input-feature values.



(c) An ℓ_2 -norm adversarial attack in a 7-class setup with increased bottleneck dimension $m = 3$. The leftmost panel shows the cosine-similarity matrix between pairs of class latent representations.

Figure 5: Visualisations of AExs in the toy model, supplementing Figure 1 of the main paper by illustrating attack mechanisms in activation and input space under varied conditions.

Table 6: Mean normalised robust accuracy ($\pm$ standard deviation across 5 seeds) for CIFAR-10 ViT models with different bottleneck dimensions (m) under ℓ_∞ -norm PGD attacks of varying perturbation magnitude (ϵ). Robust accuracy is normalised by each bottlenecked model’s clean accuracy, and the results support Sec. 4 by showing the same trend across ϵ .

ϵ	Bottleneck Dimension (m)			
	2	3	5	10
0.001	96.0% $\pm$ 0.4%	97.4% $\pm$ 0.7%	97.9% $\pm$ 0.4%	98.1% $\pm$ 0.3%
0.01	61.7% $\pm$ 3.6%	69.6% $\pm$ 3.5%	77.0% $\pm$ 0.9%	81.8% $\pm$ 3.2%
0.05	4.9% $\pm$ 1.5%	6.7% $\pm$ 1.6%	9.3% $\pm$ 0.6%	10.5% $\pm$ 0.8%
0.1	0.1% $\pm$ 0.2%	0.2% $\pm$ 0.3%	0.2% $\pm$ 0.4%	0.2% $\pm$ 0.4%
0.5	0.0% $\pm$ 0.0%	0.0% $\pm$ 0.0%	0.0% $\pm$ 0.0%	0.0% $\pm$ 0.0%

Table 7: Mean normalised robust accuracy ($\pm$ standard deviation across 5 seeds) for CIFAR-10 ViT models with different bottleneck dimensions (m) under ℓ_2 -norm PGD attacks of varying perturbation magnitude (ϵ). Robust accuracy is normalised by each bottlenecked model’s clean accuracy; the results support the main-paper findings by showing decreasing robustness for smaller m (greater superposition) and larger ϵ .

ϵ	Bottleneck Dimension (m)			
	2	3	5	10
0.1	90.4% $\pm$ 1.7%	91.8% $\pm$ 0.7%	94.6% $\pm$ 1.2%	95.0% $\pm$ 0.7%
0.5	58.5% $\pm$ 5.0%	60.8% $\pm$ 5.0%	69.2% $\pm$ 1.8%	72.6% $\pm$ 2.8%
1.0	41.7% $\pm$ 4.8%	44.0% $\pm$ 3.0%	50.4% $\pm$ 0.6%	54.9% $\pm$ 1.7%
2.0	34.0% $\pm$ 4.6%	36.2% $\pm$ 3.3%	41.5% $\pm$ 1.3%	47.4% $\pm$ 2.4%
5.0	30.2% $\pm$ 4.1%	32.9% $\pm$ 3.5%	37.3% $\pm$ 1.9%	42.7% $\pm$ 1.0%

D.2 CORRELATIONS IN CLASS FEATURES

In Sec. 3 it was correlations between inputs that drove superposition arrangements. We note that here it is not the correlations between input classes but rather the correlations in the representations at this point in the network that drive these arrangements. At the classification layer this is likely similar to the confusion matrix – *i.e.* how each class is misclassified in relation to the other classes. Classes that cluster together are those the network finds inherently similar and misclassifies together. To test this we repeat the experiment, finetuning the base ViT using timm/vit_base_patch16_384 which has been trained on ImageNet-21k (14 million images, 21,843 classes) and ImageNet (1 million images, 1,000 classes). After fine-tuning on CIFAR-10 and applying the same bottleneck training procedure as Sec. 4, the resulting geometry shows random ordering between initialisations with approximately equal spacing between features (*i.e.* neuron collapse Kothapalli (2023)). In this case performance is near 100%, meaning the confusion matrix is the identity, and the superposed loses its structure. In contrast, models trained solely on CIFAR-10 converge to similar geometries because they share the same learned difficulty structure - the same pairs of classes prove challenging to distinguish, leading to consistent superposition patterns.

D.2.1 RESNET-18

To address concerns on architectural generalisation, we conduct the same experiments using a ResNet-18 (He et al., 2016) architecture as the base model as opposed to a ViT. We achieve a slightly higher 92% clean accuracy (compared to 89% for ViT). We observe the robust accuracy falls slightly faster

Table 8: Attack transferability (%) for ℓ_∞ -norm PGD attacks on CIFAR-10 ViT models. Transferability is shown from a model trained with a specific ‘Source Seed’ (e.g., Seed 10) to three different target models, each trained with one of the seeds listed in the sub-header (e.g., ‘vs. Seeds 20/30/40’). The three slash-separated values in each cell correspond to the transferability to these three target seeds, respectively. All models within a row share the same bottleneck dimension, m . The ‘Mean $\pm$ Std’ column averages transferability across all 12 source-target seed pairings for each (ϵ, m) configuration. This supports the claim in Section 4 that higher superposition can lead to more consistent latent geometries and thus higher transferability.

ϵ	m	Seed 10 vs. Seeds 20/30/40	Seed 20 vs. Seeds 10/30/40	Seed 30 vs. Seeds 10/20/40	Seed 40 vs. Seeds 10/20/30	Mean $\pm$ Std
0.001	2	77.8/66.7/44.4	25.0/62.5/25.0	72.7/72.7/72.7	58.3/75.0/58.3	59.3 $\pm$ 17.7
	3	57.1/28.6/28.6	14.3/28.6/28.6	63.6/72.7/72.7	42.9/42.9/42.9	43.6 $\pm$ 18.4
	5	70.0/50.0/50.0	57.1/71.4/57.1	50.0/25.0/25.0	16.7/50.0/33.3	46.3 $\pm$ 16.9
	10	55.6/55.6/66.7	12.5/37.5/25.0	16.7/33.3/16.7	28.6/71.4/28.6	37.3 $\pm$ 19.3
0.01	2	60.3/52.6/48.7	63.8/42.0/60.9	53.8/40.9/52.7	51.2/55.8/46.5	52.4 $\pm$ 6.9
	3	42.9/46.4/50.0	46.4/42.9/49.1	44.2/38.9/44.2	42.9/45.1/41.8	44.6 $\pm$ 3.0
	5	43.3/41.1/43.3	39.2/30.4/41.8	46.0/40.2/47.1	41.2/35.3/32.9	40.2 $\pm$ 4.8
	10	42.5/46.0/44.8	38.0/39.4/46.5	43.5/47.8/47.8	32.6/40.0/35.8	42.1 $\pm$ 4.7
0.05	2	35.0/36.4/37.8	38.8/36.0/46.3	37.4/31.3/41.7	42.0/42.0/40.7	38.8 $\pm$ 3.8
	3	29.9/29.9/37.1	30.4/30.1/35.6	30.2/30.2/35.1	30.1/31.9/34.4	32.1 $\pm$ 2.6
	5	26.8/25.3/29.2	25.2/21.5/26.4	28.0/28.0/26.5	26.0/26.6/21.9	25.9 $\pm$ 2.2
	10	27.1/24.9/30.4	21.6/23.0/28.4	25.3/28.1/30.6	21.0/24.9/20.1	25.4 $\pm$ 3.4
0.1	2	34.2/35.6/38.2	36.7/34.9/44.5	37.9/30.0/40.8	41.2/42.0/40.3	38.0 $\pm$ 3.8
	3	29.6/28.9/35.2	28.6/29.2/34.8	29.1/28.2/34.0	28.8/31.1/32.8	30.9 $\pm$ 2.5
	5	25.4/23.3/27.0	23.8/19.6/24.6	25.9/26.7/25.4	24.6/24.3/20.7	24.3 $\pm$ 2.1
	10	24.8/22.6/29.7	20.1/21.6/26.1	22.5/25.5/28.0	19.5/23.3/18.7	23.5 $\pm$ 3.3

for ResNet-18 than the ViT at the same ϵ values. Nevertheless, we observe similar declining trends in normalised robust accuracy and increasing transferability as bottleneck pressure increases.

E MODULAR ADDITION

Sec. 5 introduces a modular addition task to investigate adversarial vulnerability in a scenario where features are orthogonal. This section provides a detailed experimental setup for this task, additional results and further discussion on how we view this sort of analysis to be useful.

E.1 LEARNED ALGORITHMIC REPRESENTATION IN MODULAR ADDITION

As detailed in Nanda et al. (2023) and Prieto et al. (2025b), the network discovers and implements a trigonometric algorithm to solve this task: the encoder projects inputs onto non basis-aligned linear representations encoding $\sin(\omega_k a)$ and $\cos(\omega_k a)$ at key frequencies $\omega_k = 2\pi k/P$. These features are represented orthogonally, *i.e.* not in superposition, and the sinusoidal values can be read off via linear probes. The subsequent MLP then applies sum-angle identities to compute $\cos((a+b)\omega_k)$ for various k , which constructively interfere at the logits to select the correct class. AExs are generated by relaxing the one-hot constraint and perturbing input $\mathbf{x}_j \in \mathbb{R}^P$ (encoding integer j) to $\mathbf{x}' = \mathbf{x} + \delta$ via ℓ_2 and ℓ_∞ -norm PGD. For simplicity, only a is attacked while b is fixed.

As explained in Nanda et al. (2023), the network learns a structured trigonometric algorithm. W_e encodes integers’ sine and cosine values for key frequencies. These features are linearly decodable sine/cosine values. Scaling the corresponding feature vector alters the encoded sinusoidal value itself, not its activation intensity. Integer embeddings $\mathbf{W}_e^j$ project onto a Fourier-like basis, creating a

Table 9: Attack transferability (%) for ℓ_2 -norm PGD attacks on CIFAR-10 ViT models. The table format, detailing source model to target model transferability, mirrors that of Tab. 8; please see its caption for a full explanation. These ℓ_2 results further support the claim in Sec. 4 that higher superposition leads to increased attack transferability.

ϵ	m	Seed 10 vs. Seeds 20/30/40	Seed 20 vs. Seeds 10/30/40	Seed 30 vs. Seeds 10/20/40	Seed 40 vs. Seeds 10/20/30	Mean $\pm$ Std
0.1	2	70.6/64.7/41.2	60.0/50.0/60.0	64.3/60.7/57.1	48.1/63.0/51.9	57.6 ± 8.0
	3	39.1/39.1/47.8	50.0/60.7/50.0	48.1/55.6/51.9	48.0/28.0/40.0	46.5 ± 8.4
	5	43.5/34.8/43.5	31.6/31.6/42.1	42.9/35.7/21.4	30.4/39.1/34.8	35.9 ± 6.4
	10	45.5/59.1/45.5	19.0/28.6/28.6	37.5/43.8/43.8	23.8/52.4/38.1	38.8 ± 11.4
0.5	2	59.6/52.1/48.9	60.8/35.4/60.8	54.1/39.4/51.4	43.8/49.5/46.7	50.2 ± 7.7
	3	38.7/40.6/48.1	36.1/36.8/41.7	37.4/36.5/43.5	34.4/34.4/36.7	38.7 ± 3.9
	5	38.4/32.0/37.6	29.4/25.7/32.1	39.1/33.9/38.3	31.8/29.9/32.7	33.4 ± 4.0
	10	38.3/37.4/37.4	26.5/28.4/31.4	32.4/35.3/35.3	23.4/28.2/25.0	31.6 ± 5.0
1.0	2	50.8/43.7/44.4	51.7/36.4/50.0	46.1/32.9/44.7	42.7/46.7/45.3	44.6 ± 5.3
	3	35.7/36.9/41.7	31.8/36.4/42.6	33.7/32.0/37.3	34.9/32.0/35.5	35.9 ± 3.3
	5	32.1/28.9/33.2	30.9/23.0/30.9	35.9/31.5/33.7	27.7/28.2/27.1	30.3 ± 3.3
	10	32.3/32.3/38.5	23.7/27.7/29.9	29.1/32.0/34.3	22.2/25.6/21.1	29.1 ± 5.0
2.0	2	47.0/41.6/44.3	48.2/36.2/48.2	43.9/33.5/42.1	43.4/43.4/42.8	42.9 ± 4.2
	3	31.9/37.7/40.3	31.1/35.6/40.6	32.6/31.1/36.8	35.4/33.3/34.3	35.1 ± 3.2
	5	29.9/25.4/32.1	26.3/22.1/26.7	32.3/31.8/29.5	27.9/27.9/23.6	28.0 ± 3.2
	10	30.8/30.4/35.7	22.5/24.9/28.6	24.8/29.7/32.7	20.4/26.1/19.0	27.1 ± 4.9
5.0	2	44.7/46.7/40.8	45.3/37.3/48.7	42.7/33.9/41.5	42.3/42.3/40.6	42.2 ± 3.8
	3	32.3/35.8/40.8	31.3/32.6/39.9	31.0/30.0/33.8	35.0/33.0/33.0	34.1 ± 3.2
	5	30.6/24.1/28.6	28.3/20.9/24.8	31.7/31.3/27.4	28.1/29.0/23.2	27.3 ± 3.3
	10	29.8/27.4/32.7	19.7/24.1/28.9	24.9/29.9/31.2	18.8/25.8/19.2	26.0 ± 4.6

circular/periodic structure in the embedding space. Unlike arrangements from data correlations, this circularity arises from the sine/cosine encoding of the (non-correlated) input integers.

Our model learns to utilise approximately seven key frequencies (ω_k). Each embedding W_e^j for an input integer x_j predominantly encodes its fourteen corresponding trigonometric features: $\cos(\omega_i x_j)$ and $\sin(\omega_i x_j)$ for $i \in \{1, 2, \dots, 7\}$. The concatenated 200-dim. vector thus provides these ≈ 28 key trigonometric features (fourteen for a and fourteen for b) to the MLP. The MLP leverages these, via sum-angle identities (e.g., $\cos(A + B) = \cos A \cos B - \sin A \sin B$), to calculate $(x_1 + x_2) \pmod{P}$. The algorithmic brittleness explored in Section 5 stems from the adversarial perturbation of these key learned trigonometric features.

E.2 EXPERIMENT DETAILS FOR MODULAR ADDITION TASK

The network is trained for addition modulo 113. Inputs are integer pairs (a, b) , $a, b \in \{0, \dots, 112\}$. Each integer is mapped to a 100-dimensional vector via a shared embedding $\mathbf{W}_E \in \mathbb{R}^{113 \times 100}$. Embeddings $\mathbf{W}_E^a$ and $\mathbf{W}_E^b$ are concatenated into a 200-dim. feature vector. This vector feeds a 3-hidden-layer MLP (200 ReLU neurons/layer). The output layer provides 113 logits for the classification. The model is trained end-to-end using cross-entropy loss and AdamW (weight decay 4), achieving 100% test accuracy. AExs are generated using PGD under ℓ_2 and ℓ_∞ norms with varying ϵ . Attacks perturb the embedding $\mathbf{W}_E^a$ (with integer b fixed) by adding a perturbation δ to cause misclassification of $(a + b) \pmod{113}$.

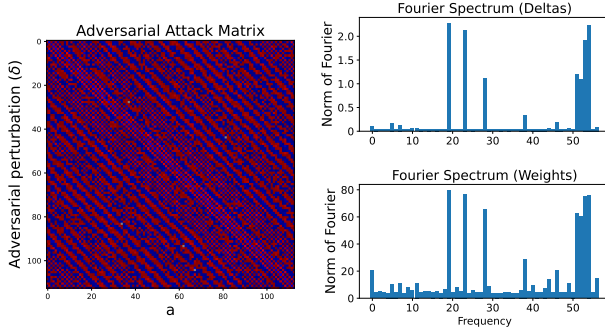


Figure 6: Left: Each column is the mean attack δ added to a to successfully attack $a + b \bmod p$. Right: The norms of the Fourier components in the δ and $W_{MLP_{out}}$.

Table 10: Comparison of minimum ℓ_2 -norm budget required to successfully attack $0 + 0 \bmod p$ between different attack types.

Attack Method	Success ϵ
PGD	0.17
Informed Attack	0.22
Single a	0.97
ℓ_∞ Noise	2.04
Uniform δ	6.20

E.3 ALGORITHMIC BRITTLINESS

The frequency-aligned perturbations cause misclassification by effectively altering the sinusoidal values read off from the encoder features. We use the hypothesised neuron-aligned case to give intuition: W_e would contain a row corresponding to the sine/cosine for each ω_k , whose element at column w_e^a contains the value $\sin(\omega_k a)$ or $\cos(\omega_k a)$. For the input a , the one-hot x_a acts to select w_e^a , giving $h_a = w_e^a$. The perturbation δ transforms one-hot x_a into a dense vector x'_a . Now $h'_a = W_e x'_a = (1 + \epsilon_a) h_a + \sum_{k \neq a} \epsilon_k w_e^k$. The introduced contributions from the $k \neq a$ sinusoidal terms corrupts the sinusoidal values, thereby disrupting the MLP’s subsequent computations and leading to misclassification. The model’s algorithm uses the constructive interference of each ω_k at the logits, and as such disrupting a single ω_k is insufficient as the other components overwhelmingly support the correct answer. Thus, effective attacks must find δ that maximally alters multiple frequency components.

E.4 ANALYTIC CONSTRUCTION OF ATTACKS AND BASELINE COMPARISONS

The main paper describes how understanding the learned trigonometric algorithm in the modular addition task allows for the analytic construction of adversarial attacks. Sec. 5 of the main paper benchmarks this ‘informed attack’ against PGD and naive baselines. This section provides further details on these baseline attack methods. The baselines are:

- **PGD Attack:** Standard PGD used to find the adversary.
- **Informed Attack:** The analytically constructed perturbation, from combinations of cosines.
- **Single a :** This involved finding the minimal attack to a single input to cause misclassification.
- **ℓ_∞ Noise:** This baseline involved adding random noise δ_{noise} to the input a . Each component of δ_{noise} was sampled independently and uniformly from $[-\epsilon, \epsilon]$. The ϵ that successfully induced a misclassification is reported.
- **Uniform δ :** The perturbation vector δ is constrained to have all components equal, i.e., $\delta_j = c \cdot 1$, where c is a scalar. The magnitude of c is reported such that adding δ to the input causes a misclassification.

E.5 DETAILS ON CERTIFIED ROBUST TRAINING

To investigate the impact of defences on the identified vulnerability, we performed certified robust training. This aimed to make the network provably robust against ℓ_∞ -norm perturbations. The certified training method employed was an abstraction-based technique (Wong & Kolter, 2018; Raghunathan et al., 2018; Palma et al., 2022), namely RSIP-IBP (Zhang et al., 2020; Henriksen & Lomuscio, 2023). RSIP-IBP utilises Interval Bound Propagation (IBP) (Katz et al., 2017; Ehlers, 2017) and Reversed Symbolic Interval Propagation (RSIP) to compute reachable output sets for network layers, thereby enabling formal certification of robustness within a given perturbation budget. The training loss function incorporated both a standard loss component and an RSIP loss component. Certified training was conducted for 1000 epochs, with the initial 200 epochs serving as a warm-up phase. The learning rate was 1×10^{-3} and weight decay 1×10^{-4} . The same train/test data splits

were used as for the standard model training. The results of certified training across different training perturbation budgets (ϵ_{train}) and evaluated against various attack perturbation budgets (ϵ_{attack}) are summarised in Tab. 11.

Table 11: Verified robust accuracy (%) for modular addition models under ℓ_∞ -norm perturbations. The table compares a standard (pre-robustification) model with models certifiably trained using RSIP-IBP with different training perturbation budgets (ϵ_{train}). Accuracies are evaluated against various attack budgets (ϵ_{attack}). These results support the discussion in Section 5.1 of the main paper, indicating that certified training improved robustness to specified ϵ_{train} but attacks still succeeded by exploiting the same algorithmic sensitivities at higher ϵ_{attack} .

Attack Epsilon (ϵ_{attack})	Train Epsilon (ϵ_{train})	Standard Acc. (Pre-Rob., %)	Robust Acc. (Pre-Rob., %)	Standard Acc. (Robust, %)	Robust Acc. (Robust, %)
5.0×10^{-4}	10^{-4}	100.0	100.0	100.0	100.0
	10^{-3}				100.0
1.0×10^{-3}	10^{-4}	100.0	76.0	100.0	93.8
	10^{-3}				100.0
2.0×10^{-3}	10^{-4}	100.0	0.0	100.0	0.0
	10^{-3}				95.7
5.0×10^{-3}	10^{-4}	100.0	0.0	100.0	0.0
	10^{-3}				85.3

As discussed in Sec. 5, subsequent PGD attacks on these robustly trained models retained the same qualitative mechanism: successful perturbations continued to exploit the key frequency components inherent in the learned algorithm. The primary effect of certified training was therefore to increase the perturbation magnitude required for misclassification at epsilons for which certification succeeded, rather than to fundamentally alter the attack vector’s alignment with the model’s algorithmic sensitivities.

E.6 ON A POTENTIAL RECONCILIATION WITH SUPERPOSITION VULNERABILITY

Although the vulnerability examined here arises without feature superposition, it exhibits a notable phenomenological resemblance to superposition-driven vulnerabilities: both depend on sparsity in the input representation. This motivates our *reconciliation conjecture*: algorithmic vulnerabilities may persist in networks with orthogonal representations when upstream sparsity assumptions are violated. If earlier layers produce noisy or non-sparse features, potentially through adversarial interference, those deviations can propagate into otherwise well-isolated downstream circuits. Circuits that operate correctly on clean inputs may then fail because their effective inputs are already perturbed. This provides a plausible route by which algorithm-disruption mechanisms, such as the frequency attacks studied here, could be activated even without local superposition. More broadly, the interaction between upstream superposition and downstream algorithmic brittleness, as well as layer-by-layer characterisation of the specific vulnerability mechanisms triggered by AExs, remains underexplored. We leave a detailed investigation grounded in empirical evidence and theoretical modelling as a promising direction for future work.

F INSPECTING ADVERSARIAL PERTURBATIONS VIA SAE FEATURES

The main-paper analyses focus on scenarios with either explicitly defined class features (the toy model) or class-level representations in a bottleneck (the ViT experiments). In large practical networks, however, interference is likely to occur among unlabelled latent features distributed across layers. As noted in the future-work discussion, SAEs provide a promising unsupervised method for extracting more linear features from NNs (Bricken et al., 2023). This appendix presents a preliminary investigation of whether SAE-extracted features can characterise how a large-scale ViT responds internally to AExs relative to clean inputs. These initial results are intended to assess the feasibility of future feature-level analyses of adversarial phenomena in large models, rather than to establish a

conclusive new set of findings. We first describe the experimental setup and validate SAE behaviour on adversarial inputs, then present exploratory layer-wise feature-difference analyses.

F.1 EXPERIMENT SETUP

We analyse SAE-extracted features from the vision encoder of LAION/CLIP-ViT-B-32-DataComp.XL-s13B-b90K, a 12-layer ViT with $\approx 70\%$ zero-shot accuracy on ImageNet-1k Deng et al. (2009). We use open-source Prisma SAEs that map the 768-dimensional activation space to a dictionary of 49,152 features ($64\times$ expansion) (Joseph et al., 2025). Across all layers, we examine SAEs trained on either mean patch-token representations or CLS-token representations. We analyse features activated above a high threshold (0.1), to capture changes in the strongest activations, and above a low threshold (0.001). For the SAE comparisons, we generate 10,000 original/successfully attacked image pairs under ℓ_∞ -norm attacks with $\epsilon = 0.02$. This setup provides an initial probe of how adversarial perturbations may manifest at the level of SAE features.

F.2 SAE VALIDITY FOR ADVERSARIAL INPUTS

Because AExs are out-of-distribution inputs, and because we aim to use standardly trained SAEs to analyse them, we first perform a basic validation. This subsection examines how inserting SAEs at inference time affects model accuracy and reconstruction for both clean and adversarial inputs. Before interpreting differences in feature activations, we therefore assess whether the SAEs continue to process features relevant to classification even under adversarial conditions.

Table 12: Impact on ImageNet-1k classification accuracy when inserting a pre-trained SAE (one layer at a time) during inference for original, ℓ_∞ adversarial ($\epsilon = 0.02$), and random noisy inputs. ‘Accuracy’ refers to the ViT’s top-1 accuracy. ‘Original input + SAE’ means the clean ViT activations at a given layer are passed through the SAE then its reconstruction is passed to the next layer. The slight ‘de-attacking’ effect (improved accuracy for ‘Adv. input + SAE’ vs ‘Adv. input’) suggests SAEs might filter some adversarial noise, but overall, the model still largely misclassifies, indicating salient adversarial changes for misclassification are processed.

Metric	Layer 2	Layer 5	Layer 8	Layer 11
Accuracy				
Original input	1.00	1.00	1.00	1.00
Original input + SAE	0.99	0.99	0.99	0.86
Adv. input	0.00	0.00	0.00	0.00
Adv. input + SAE	0.02	0.01	0.01	0.11
Noisy input	0.98	0.98	0.98	0.98
Noisy input + SAE	0.98	0.98	0.98	0.85
SAE Improvement				
Original input	-0.01	-0.01	-0.01	-0.14
Adv. input	0.02	0.01	0.01	0.11
Noisy input	-0.00	-0.00	-0.00	-0.13

F.3 LAYER-WISE SAE FEATURE DIFFERENCES

This subsection presents exploratory visualisations comparing SAE feature activations across ViT layers for original, ℓ_∞ adversarial, and random noisy inputs. These visualisations are intended to identify promising directions for more detailed future investigation into how adversarial attacks distinctively alter sparse feature representations. We examine:

- **Activation count:** Histograms comparing the number of SAE features activated above a set threshold in a layer for original, adversarial, and noisy images. This explores whether attacks characteristically alter feature sparsity.

- Feature overlap: The degree of overlap (number of common activated features) between (original vs. adversarial) and (original vs. noisy) conditions. This explores if attacks activate a distinctly different set of features.
- Jaccard Similarity: The Jaccard similarity between sets of activated features, offering a normalised measure of overlap.
- Mean activation: Histograms of mean activation values for features active above a threshold. This looks for changes in the intensity of (key) features.
- Distinct Features: The count of unique features activated across an image set for each condition.

F.4 CLS TOKEN SAE FEATURES WITH HIGH ACTIVATION THRESHOLD (0.1)

This subsection presents a preliminary analysis of SAE features derived from CLS tokens, restricting attention to features whose activation exceeds 0.1. Figures 7 through 10 compare activation-count histograms, feature overlap, Jaccard similarity, and mean-activation histograms for original, adversarial, and random noisy inputs across ViT layers.

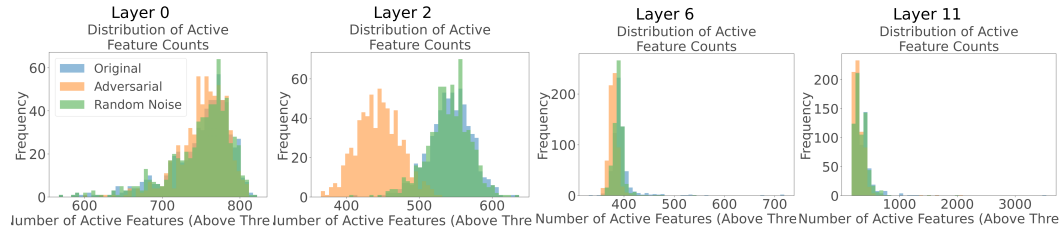


Figure 7: Histogram of activation counts for CLS token SAE features (threshold 0.1).

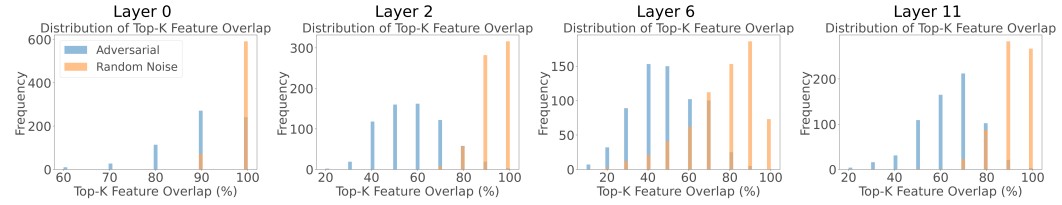


Figure 8: Feature overlap between original vs. attacked and original vs. noisy images for CLS token SAE features (threshold 0.1).

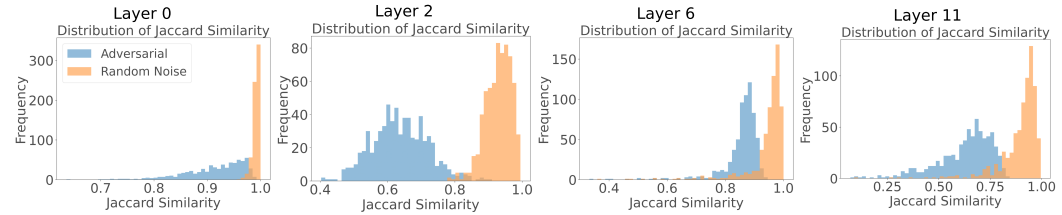


Figure 9: Comparison of Jaccard similarity for CLS token SAE features (threshold 0.1).

F.5 CLS TOKEN SAE FEATURES WITH LOW ACTIVATION THRESHOLD (0.001)

Here, we examine SAE features from CLS tokens using the lower activation threshold 0.001, so that the analysis is not restricted to only the most strongly activated features. The following figures show activation-count histograms (Fig. 11), feature overlap (Fig. 12), Jaccard similarity (Fig. 13), and mean-activation histograms (Fig. 14) across layers for original, adversarial, and noisy inputs.

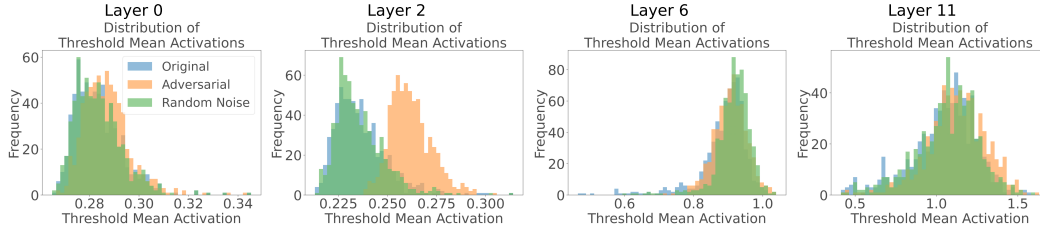


Figure 10: Histogram of mean thresholds for CLS token SAE features (threshold 0.1).

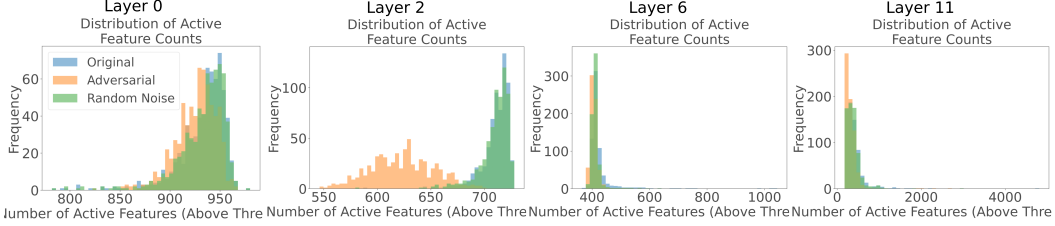


Figure 11: Histogram of activation counts for CLS token SAE features (threshold 0.001), with noise.

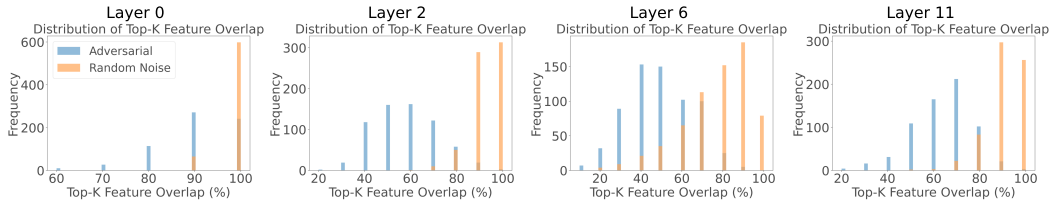


Figure 12: Feature overlap between original vs. attacked and original vs. noisy images for CLS token SAE features (threshold 0.001).

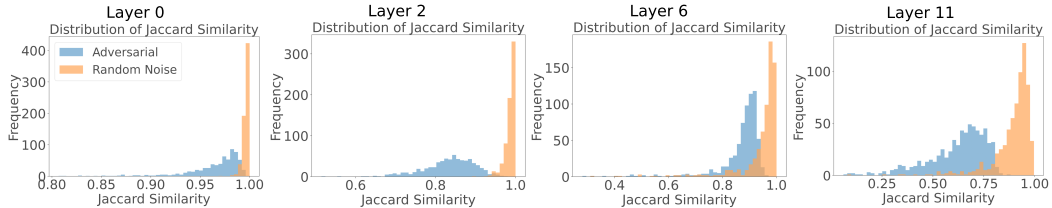


Figure 13: Comparison of Jaccard similarity for CLS token SAE features (threshold 0.001).

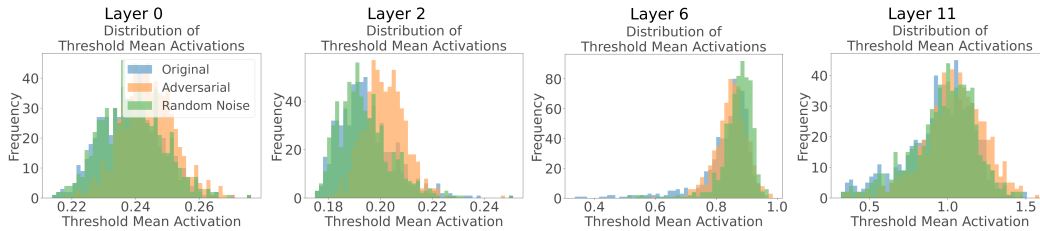


Figure 14: Histogram of mean activations for CLS token SAE features (threshold 0.001), with noise.

F.6 PATCH TOKEN SAE FEATURES WITH HIGH ACTIVATION THRESHOLD (0.1)

This subsection shifts attention to SAE features derived from the mean of patch tokens, again using the high activation threshold 0.1. We compare activation-count histograms (Fig. 15), distinct-feature-

count histograms (Fig. 16), feature overlap (Fig. 17), Jaccard similarity (Fig. 18), and mean-activation histograms (Fig. 19) for original, adversarial, and noisy inputs.

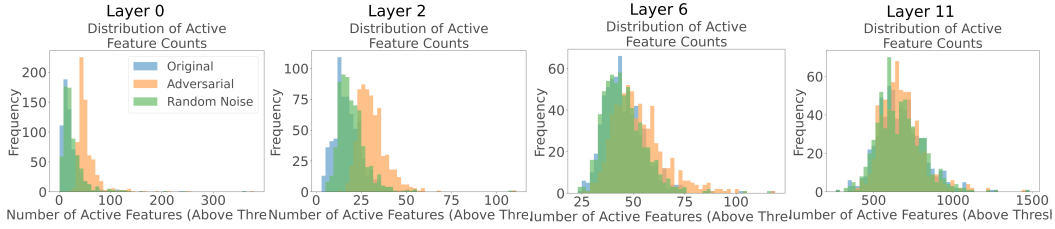


Figure 15: Histogram of activation counts for patch token SAE features (threshold 0.1), with noise.

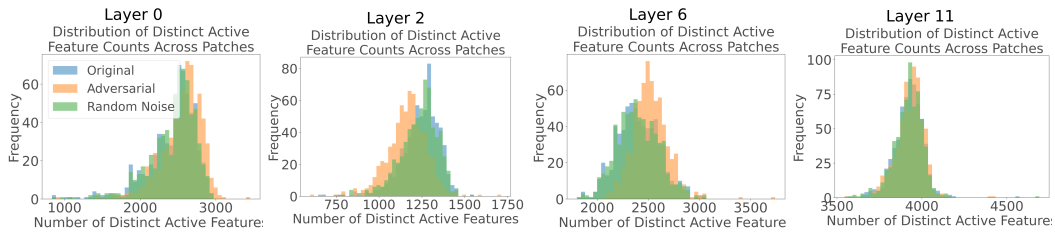


Figure 16: Histogram of distinct patch features for patch token SAE features (threshold 0.1), with noise.

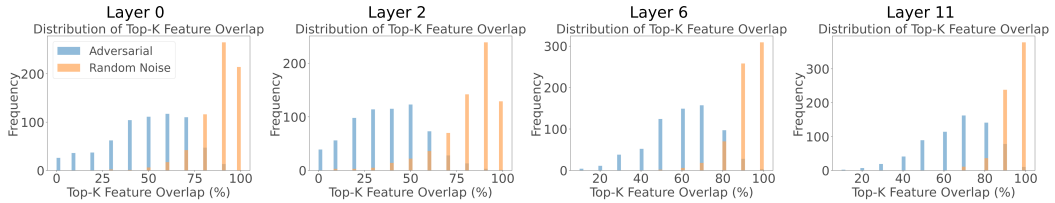


Figure 17: Feature overlap between original vs. attacked and original vs. noisy images for patch token SAE features (threshold 0.1).

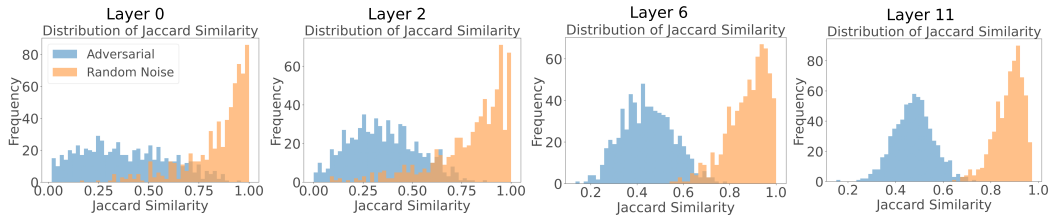


Figure 18: Comparison of Jaccard similarity for patch token SAE features (threshold 0.1).

F.7 PATCH TOKEN SAE FEATURES WITH LOW ACTIVATION THRESHOLD (0.001)

Finally, this subsection examines SAE features from patch-token means under the lower activation threshold 0.001. The figures report activation-count histograms (Fig. 20), distinct-feature counts

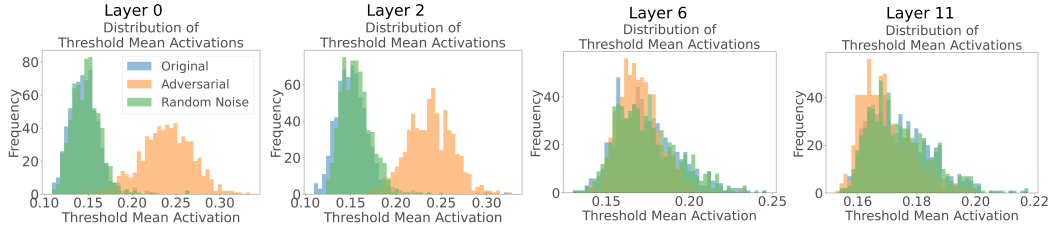


Figure 19: Histogram of mean thresholds for patch token SAE features (threshold 0.1), with noise.

(Fig. 21), feature overlap (Fig. 22), Jaccard similarity (Fig. 23), and mean-activation histograms (Fig. 24) for original, adversarial, and noisy inputs.

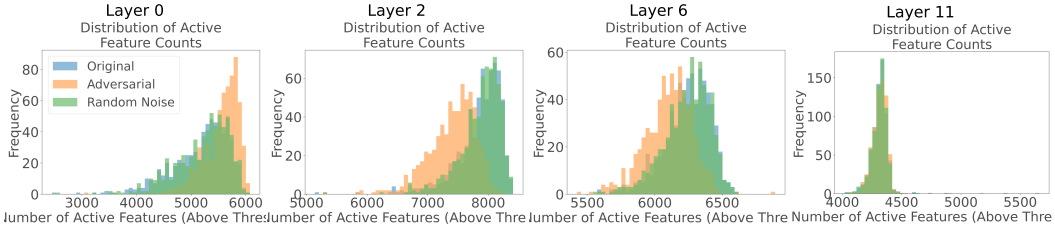


Figure 20: Histogram of activation counts for patch token SAE features (threshold 0.001), with noise.

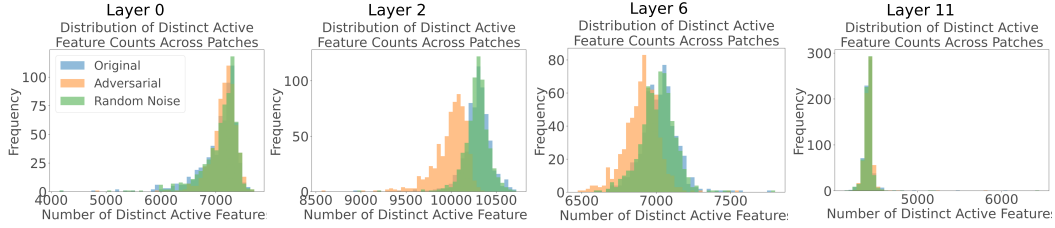


Figure 21: Histogram of distinct patch features for patch token SAE features (threshold 0.001), with noise.

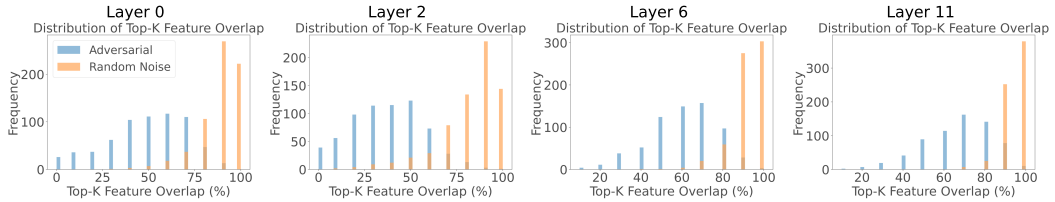


Figure 22: Feature overlap between original vs. attacked and original vs. noisy images for patch token SAE features (threshold 0.001).

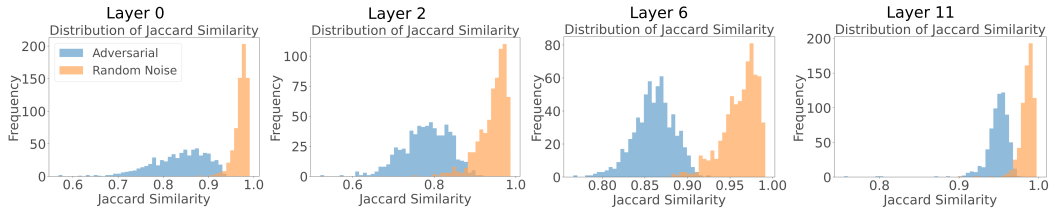


Figure 23: Comparison of Jaccard similarity for patch token SAE features (threshold 0.001).

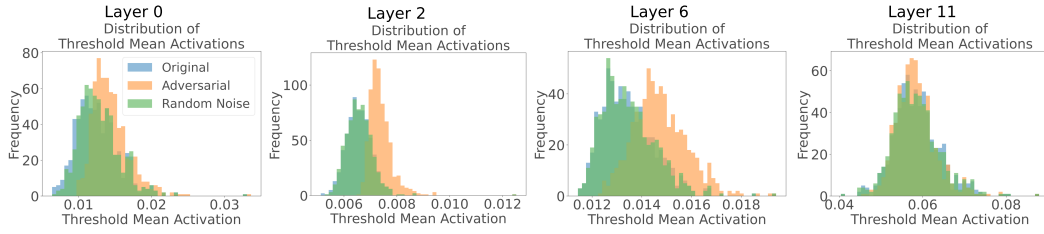


Figure 24: Histogram of mean activations for patch token SAE features (threshold 0.001), with noise.

G ACRONYMS

AEx adversarial example

CE cross entropy

LRH linear representation hypothesis

MLP Multilayer Perceptron

MSE mean squared error

NN neural network

PGD projected gradient descent

SAE sparse autoencoder

ViT vision transformer